① THEME 1 : ITEM 1 (WAKISSHA)

(a)(ii) Meaning of refractive index of a substance.

The refractive index of a substance is the ratio of the speed of light in vacuum (or air) to the speed of light in that substance.

$$n = \frac{c}{v}$$

where n = refractive index
c = speed of light in vacuum
v = speed of light in the substance.

Causes of refraction.

1. A change in the speed of light when it passes from one medium to another.
2. The two media have different optical densities (refractive indices).

The green gelatin layer acts as a colour filter. It allows mainly green light to pass through and absorbs most of the other colours. Therefore objects seen from inside appear different from those seen from outside.

(iii) Which type of glass should be taken for use.

Given: speed of light in vaccum, $c = 3.0 \times 10^8\ ms^{-1}$

Glass	Speed of light (v)	Refractive index $n = \frac{c}{v}$	Range required
A	$1.97 \times 10^8\ ms^{-1}$	$n = \frac{3.0 \times 10^8}{1.97 \times 10^8} = 1.52$	1.50 – 1.70
B	$2.14 \times 10^8\ ms^{-1}$	$n = \frac{3.0 \times 10^8}{2.14 \times 10^8} = 1.40$	

Glass A has $n \approx 1.52$ which lies in the range 1.50 – 1.70.
∴ Glass A should be selected.

(iii) Which type of lens should be selected for security cameras.

Power of lens, $P = \frac{1}{f}$ (f in metres)

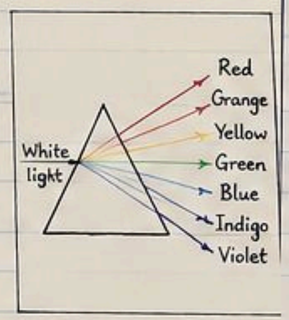
Lens	Power (P)	Focal length, $f = \frac{1}{P}$	Focal length in cm	Range required
A	8.3 D	$f = \frac{1}{8.3} = 0.1205$ m	= 12.05 cm	10.0 – 15.0 cm
B	5.0 D	$f = \frac{1}{5.0} = 0.20$ m	= 20 cm	

Lens A has focal length ≈ 12.05 cm which is within the required range.
∴ Lens A should be selected.

(b) Meaning of dispersion and why objects appear different.

Dispersion is the splitting of white light into its constituent colours when it passes through a transparent medium.

White light consists of different colours (wavelengths). These colours travel at different speeds in glass and are therefore refracted through different angles.



② THEME 1 : ITEM 1 (JJEB)

(a)(ii) Spectacles.

Given: object distance, $u = 20$ cm
image distance, $v = 30$ cm

Using the lens formula,

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$
$$= \frac{1}{20} + \frac{1}{30}$$
$$= \frac{3+2}{60}$$
$$= \frac{5}{60} = \frac{1}{12}$$

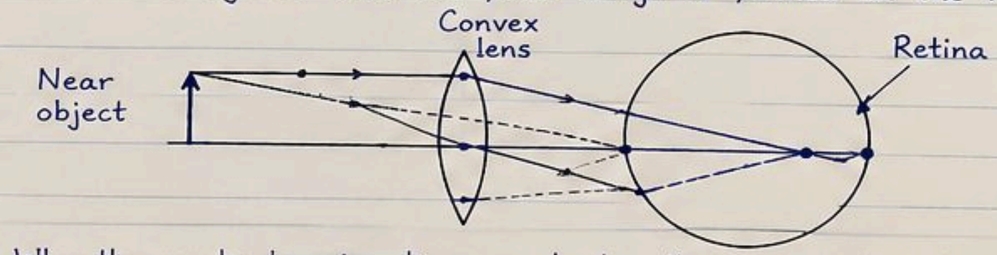
∴ $f = 12$ cm $= 0.12$ m

Power, $P = \frac{1}{f} = \frac{1}{0.12} = 8.33$ D

The recommended power is not less than + 8.29 D.
Since 8.33 D > 8.29 D, the spectacles can work for Jennie.
The power is positive, therefore the lens is CONVEX (converging).

Eye defect and how it is corrected.

Jennie suffers from LONG-SIGHTEDNESS (HYPERMETROPIA).
In hypermetropia, the image of a near object would be formed behind the retina. A convex lens converges the rays before they enter the eye so that the final image is formed on the retina.



Why the monkey's noise disappeared when the pipe was turned on.

Wavelength of sound, $\lambda = 0.2$ m
Speed of sound, $v = 340\ ms^{-1}$
Frequency, $f = \frac{v}{\lambda} = \frac{340}{0.2} =$ **1700 Hz**

When the pipe is turned on, it produces sound of a frequency close to the natural frequency of the air column. The air column undergoes RESONANCE, producing strong vibrations. These vibrations interfere with the monkey's sound, therefore the monkey's sound is greatly reduced and finally becomes inaudible.

RESONANCE is the large amplitude vibration of a system when it is forced to vibrate at, or very close to, its natural frequency.

3. THEME 1 : ITEM 1 (UMTA)

(a) Why the beam failed to clearly illuminate the ground boundary areas after replacing with wide-aperture concave mirror.

- A concave mirror produces a parallel (collimated) beam when the light source is placed at its focus.
- With a narrow aperture, only rays near the principal axis are used. These rays can be reflected nearly parallel forming a narrow and. intense beam which reaches far and clearly illuminates the boundary.
- With a wide aperture, rays from the outer parts of the mirror are used. These rays do not become parallel; they either diverge or converge at different directions after reflection.
- Hence the beam spreads out (diverges), becomes weak and fails to clearly illuminate the ground boundary areas.

(b) The suitable warning light for the top of the tower.

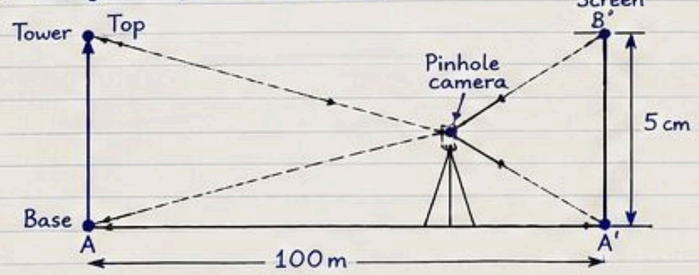
A flashing red aviation warning light (high intensity LED beacon) with steady or flash mode.

Reason:

- Red colour is least scattered by air and fog hence easily seen from far.
- Flashing light attracts attention even in day and night conditions.

(c) Determining the height of the watchtower.

(ii) Using the pinhole camera.



Let the actual height of the tower = H

Image height on the screen, h_i = 5 cm = 0.05 m

Distance of camera from tower, u = 100 m

Using similar triangles,

$$\frac{H}{u} = \frac{h_i}{v}$$

For pinhole camera, the image is formed on the screen at $v \approx u = 100$ m.

$$\therefore H = \frac{h_i \times u}{v} = \frac{0.05 \times 100}{100} = 0.05 \text{ m} \times 100 = \boxed{5.0 \text{ m}}$$

Height of the tower, H = 5.0 m

(iii) Using the echo method.

Echo time, t = 0.14 s

Speed of sound in air, $v = 340\,m\,s^{-1}$

The sound travels to the top of the tower and back.

$\therefore$ Total distance = 2H

Hence,

$$2H = v \times t$$

$$H = \frac{vt}{2} = \frac{340 \times 0.14}{2}$$

$$= \frac{47.6}{2} = 23.8 \text{ m}$$

Height, H = 23.8 m

(iii) Comparing the two results.

Pinhole camera result = 5.0 m

Echo method result = 23.8 m

The echo method is more accurate.

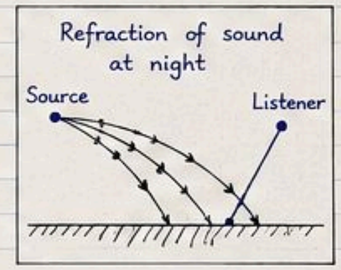
Reason: Echo method is not affected by alignment and measurement errors that may occur in the pinhole camera method.

4. THEME 1 : ITEM 1 (KAMSSA)

(a)(ii) Why did they hear the sound of the gunshot after some time?

Sound travels with a finite speed of about $340\,m\,s^{-1}$ in air. Therefore it takes some time to reach the observers. (Light travels extremely fast and the flash is seen almost instantly).

(a)(iii) Why can sound travel at night than in the day?



- At night the air near the ground is cool while the air above is warmer.
- Sound bends towards the cooler (denser) layers near the ground (refraction of sound).
- Hence sound is carried over longer distances and can be heard further. In daytime the air near the ground is warm while above is cooler.
- Sound bends away from the ground into cooler layers and is not carried far.
- Hence sound does not travel far in the day.

(a)(iii) Distance between the two cliffs.

Given:

Speed of sound, $v = 340\,m\,s^{-1}$

Time interval between first and second echo, $\Delta t = 3.5$ s

Echoes are heard successively from the nearer cliff and then the farther cliff.

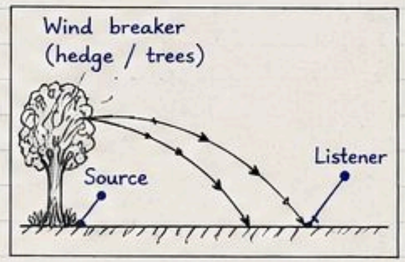
Extra distance travelled by the sound = Distance between the cliffs (d).

Total extra time = 3.5 s

$$\therefore d = v \times \Delta t = 340 \times 3.5 = 1190 \text{ m}$$

Distance between the two cliffs = 1190 m

(b) Why did the guides advise students to carry wind breakers?



- Refraction of sound at night makes the sound bend towards the ground.
- Wind breakers reduce wind and mixing of air layers.
- This keeps the air near the ground cool and stable, enhancing downward refraction of sound.
- Hence sound can be heard more clearly and over greater distances.

(c) Calculation of the critical angle for the rock material.

Given: Refractive index of the rock, n = 1.55

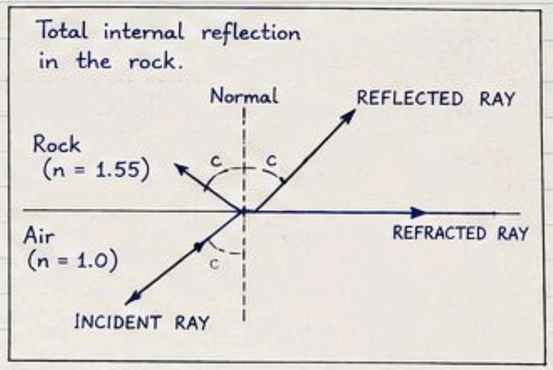
For critical angle, $\sin c = \frac{1}{n}$

$$c = \sin^{-1}\left(\frac{1}{1.55}\right)$$

$$c = \sin^{-1}(0.645)$$

$$c \approx 40.2^\circ$$

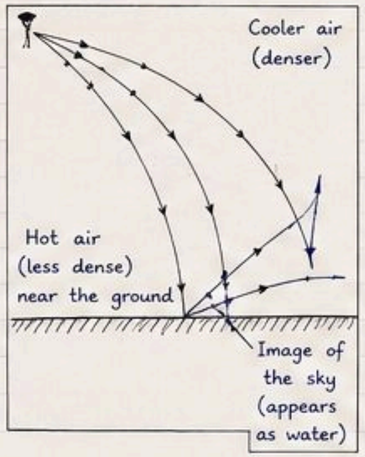
Critical angle, $c \approx 40.2^\circ$



5. THEME 1 : ITEM 1 (UTEC)

(a)(ii) Explain how the mirage was formed.

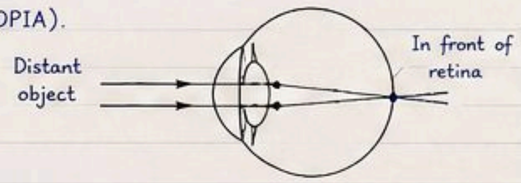
- During the day, the ground surface is strongly heated by the sun.
- The air in contact with the hot ground becomes hot and expands and is therefore less dense.
- Above it there is cooler and denser air.
- Light from the sky travels from cooler (denser) air to the hotter (less dense) air near the ground.
- It is refracted continuously away from the normal.
- When the angle of incidence becomes greater than the critical angle, the light is totally internally reflected back to the eye.
- The eye receives the rays as if they came from the ground. Therefore, the sky appears as water on the road. This phenomenon is called a MIRAGE.



(a)(ii) Identify the eye defect raised and explain.

The girl is suffering from SHORT-SIGHTEDNESS (MYOPIA).

- She is unable to see distant objects clearly.
- In myopia, rays from a distant object are focused in front of the retina instead of on it.



(a)(iii) State the effect of excessive UV radiation.

- It can cause damage to the skin (e.g. sunburn and skin cancer).
- It can cause damage to the eyes such as cataracts.
- It can weaken the immune system.

(b) The echo is used to determine the width of a valley.

Given: Echo time, $t = 5\,s$

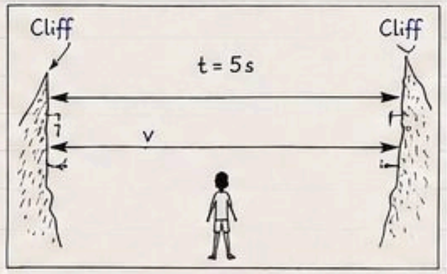
Speed of sound in air, $v = 340\,m\,s^{-1}$

The echo time is for the sound to travel to the cliff and back.

Therefore, total distance = 2 x width of valley.

Width of valley, $w = \frac{vt}{2} = \frac{340 \times 5}{2} = 850\,m$

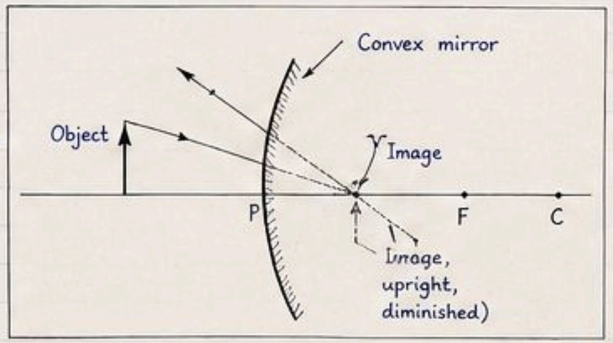
Width of the valley = 850 m



(c) State any one use of the convex mirror.

- Used as a rear view/side mirror in vehicles.
- Used as a security mirror in shops and at bends on the road.

(Any one correct use)



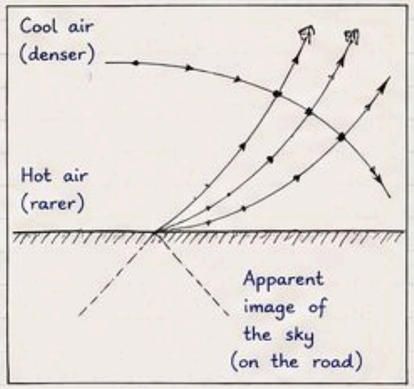
6. THEME 1 : ITEM 1 (WENSSEC / MASAKA)

(a)(ii) Explain why the daytime sky appears blue.

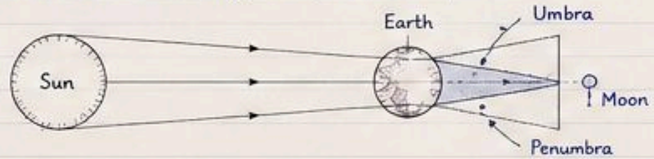
- Sunlight is made up of different colours.
- When the sunlight enters the atmosphere, it is scattered by tiny particles of dust and air molecules.
- Blue light has the shortest wavelength and is scattered the most.
- Therefore, blue light comes from all directions and the sky appears blue.

(ii) How is a mirage formed?

- The ground surface is heated by the sun.
- The air near the ground becomes hot and less dense while the air above is cooler and denser.
- Light from the sky passes from denser to rarer air and is refracted away from the normal.
- When the angle of incidence becomes greater than the critical angle, total internal reflection occurs.
- The reflected rays enter the eye which sees the sky as if it were on the ground.



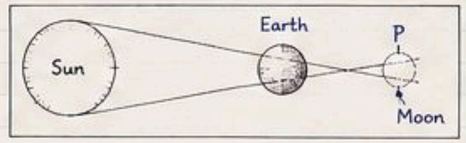
(b)(ii) Draw a well labelled diagram of a lunar eclipse.



(iii) State what would happen if the Moon were at position P.

There would be no lunar eclipse.

(The Moon would be outside the Earth's shadow so it would not enter the umbra.)



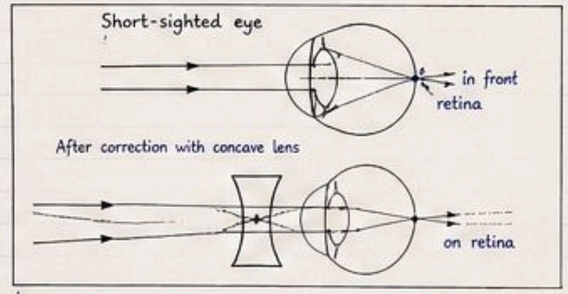
(c)(ii) Name the magnifying instrument that the Moon visitor used.

The instrument is a MICROSCOPE.

(ii) The visitor is short sighted as well.

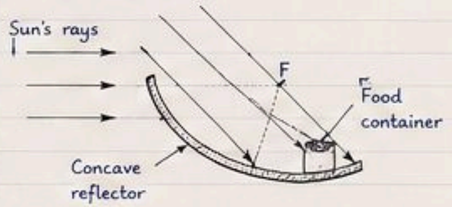
Explain how this defect can be corrected.

- It is corrected by using a concave (diverging) lens of suitable power.
- The lens diverges the incoming rays so that the eye lens focuses the image on the retina.

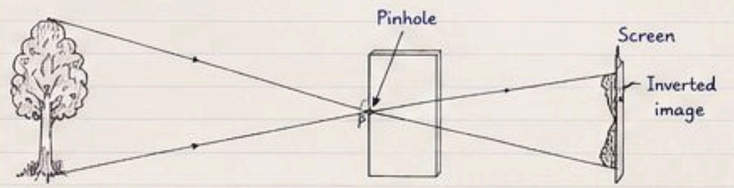


(d) Use ray diagrams to show how the solar cooker and the pinhole camera work.

(ii) Solar cooker



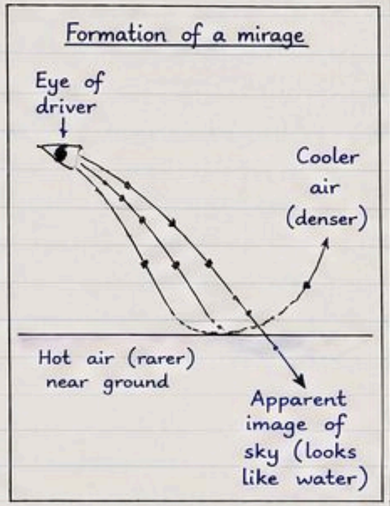
(ii) Pinhole camera



5. THEME 1 : ITEM 1 (UTEC)

(a)(i) Why did the driver think he was near water while it was a mirage?

- On a hot day, the layer of air near the ground is very hot and less dense while the air higher up is cooler and denser.
- Light from the blue sky travels from denser air (above) to rarer air (near the ground).
- It is refracted away from the normal and when the angle of incidence exceeds the critical angle, it undergoes total internal reflection.
- The eye receives the light as if it came from the ground.
- The driver sees an inverted image of the sky on the ground which looks like water.



(a)(iii) Which eye defect was being suffered from by the driver and the cause.

The driver was suffering from SHORT-SIGHTEDNESS (MYOPIA).

Cause: The eyeball is elongated or the eye lens is too curved (highly converging) so that rays from distant objects are focused in front of the retina.

(a)(iii) Give any two harmful effects of ultraviolet radiation to humans.

- Causes sunburn (reddening and pain of the skin).
- Causes skin cancer.

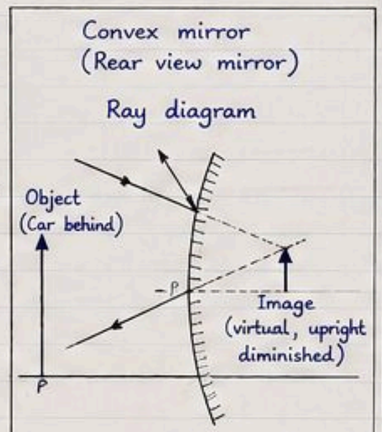
(Other correct effects: damages eyes causing cataracts, premature ageing of skin, weakens immune system, etc.)

(b)(i) Why did the doctor not see an echo for his voice but heard the birds' echoes?

- His voice is of low frequency (long wavelength) and is absorbed by the surroundings.
- High frequency sounds (like those of birds) are not easily absorbed and are reflected to produce echoes.
- Therefore his voice did not produce an echo while the birds' sounds did.

(b)(iii) How was the truck driver able to see the driver behind even though the rear view mirror is curved?

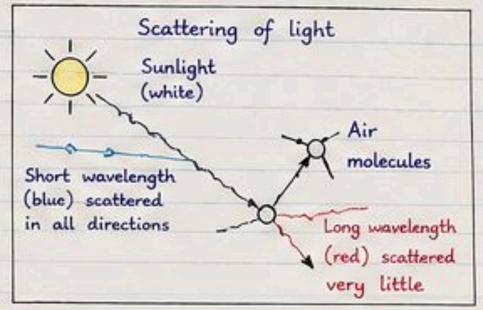
The rear view mirror is a CONVEX mirror. Convex mirrors always form upright, virtual and diminished images.
They also give a wide field of view; hence the driver can see the driver behind.



6. THEME 1 : ITEM 1 (WENSSEC/MASAKA)

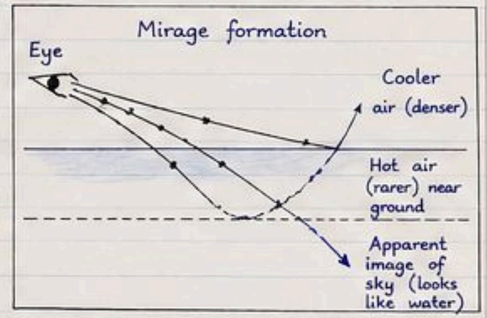
(a)(ii) Why does the blue sky appear blue?

- White light from the sun is made up of different colours.
- Air molecules scatter the shorter wavelengths (blue/violet) much more than the longer wavelengths (red).
- Blue light is scattered in all directions and enters our eyes from all parts of the sky; hence the sky appears blue.



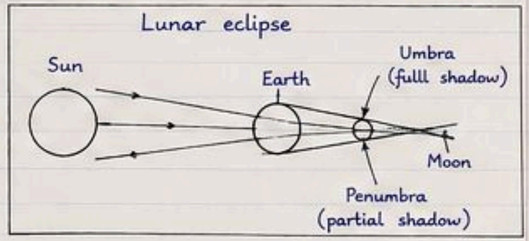
(a)(ii) Why does the mirage appear? (2 reasons)

- On a hot day the air near the ground is hot and less dense while the air above is cooler and denser.
- Light from the sky is refracted away from the normal as it travels from denser to rarer layers.
- When the angle of incidence is greater than the critical angle, total internal reflection occurs.
- The eye receives this light as if it came from the ground, so an inverted image of the sky is seen on the ground which looks like water (mirage).



(a)(iii) What is this lunar event called and how does it occur?

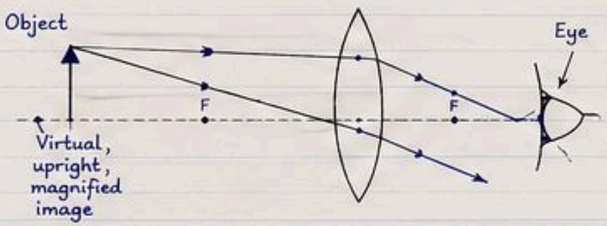
This event is a LUNAR ECLIPSE.
It occurs when the Earth comes between the Sun and the Moon. The Earth's shadow falls on the Moon and prevents sunlight from reaching the Moon.



(b)(i) Identify the instrument.

The instrument is a SIMPLE MAGNIFYING GLASS.
(Converging/convex lens used as a magnifier.)

(b)(iii) Draw a labelled ray diagram.

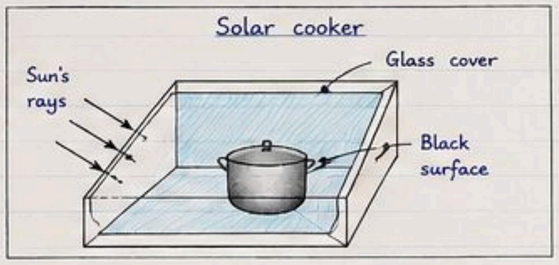


(b)(iii) Which eye defect is being corrected? How is it corrected?

Eye defect: LONG-SIGHTEDNESS (HYPERMETROPIA).
Correction: By using a CONVEX (converging) lens which converges light before it enters the eye so that the image is formed on the retina.

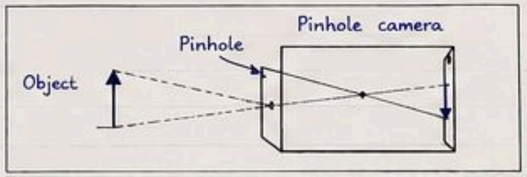
(c)(i) How does the solar cooker work?

- The black surface absorbs solar radiation and gets hot.
- The glass cover allows short wave radiation in but reduces loss by convection and re-radiation.
- Heat is trapped inside (greenhouse effect) and cooks the food.



(c)(ii) What is the use of the pinhole camera?

It is used to form an inverted real image of an object on a screen.
It is used in astronomy to observe eclipses safely.



THEME 2: RADIOACTIVITY AND NUCLEAR PHYSICS

1. ITEM 2 [WAKISSHA]

The table shows the relationship between thickness of concrete and the radiation intensity that penetrates it.

Thickness of concrete (cm)	0	10	20	30	40
Radiation intensity (units)	400	120	50	15	5

(a) Safe thickness of concrete.

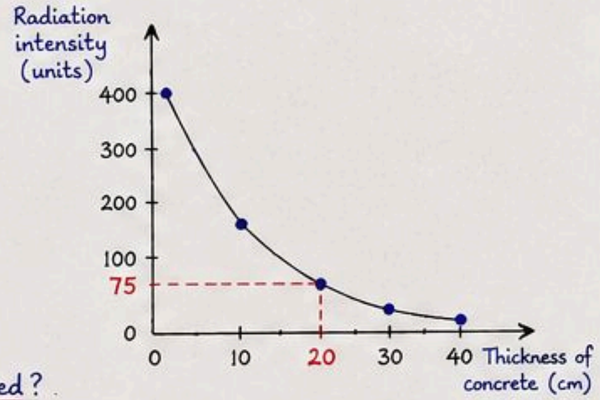
Radiation intensity above 75 units is unsafe.

From the table:

At 10 cm, intensity = 120 units (unsafe)

At 20 cm, intensity = 50 units (safe)

∴ Safe thickness = **20 cm**



(b) Why does the bunker have to be constructed?

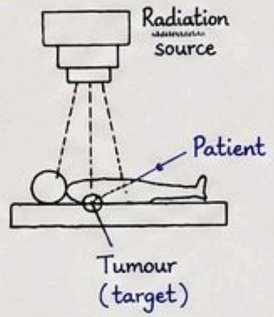
Radioactive substances emit ionising radiation which can penetrate living tissues and damage cells.

A bunker made of thick concrete absorbs/attenuates the radiation. This reduces the intensity of radiation reaching workers and members of the public.

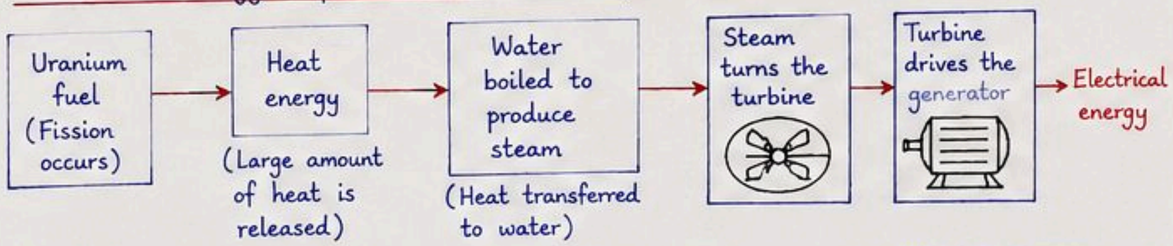
Therefore the risk of radiation effects such as cell damage, mutations and cancer is reduced. The radioactive source is thus kept in a controlled and well shielded area.

(c) How radioactive radiation is used to treat cancer.

A radioactive source is directed at cancerous tissue (tumour). The ionising radiation damages the DNA of cancer cells and destroys them or prevents them from multiplying. This controlled use of radiation is called radiotherapy.



(d) How nuclear energy is produced in a nuclear power plant.



Energy conversion: Nuclear energy → Heat energy → Kinetic energy → Electrical energy.

2. ITEM 2 [JJEB]

A radioactive mineral element (half-life = 2 years) is installed in a nuclear reactor.

Initial count rate = 1000 counts per minute.

Background radiation = 45 counts per minute.

Electricity production should start when the count rate of the element is 10 counts per minute above the background.

Required total count rate = 45 + 10 = 55 counts per minute.

(a) (i) Time when electricity production will start.

Required count rate of the element = 10 counts per minute.

Using $A = A_0 \left(\frac{1}{2}\right)^n$

$10 = 1000 \left(\frac{1}{2}\right)^n$

$\frac{10}{1000} = \left(\frac{1}{2}\right)^n$

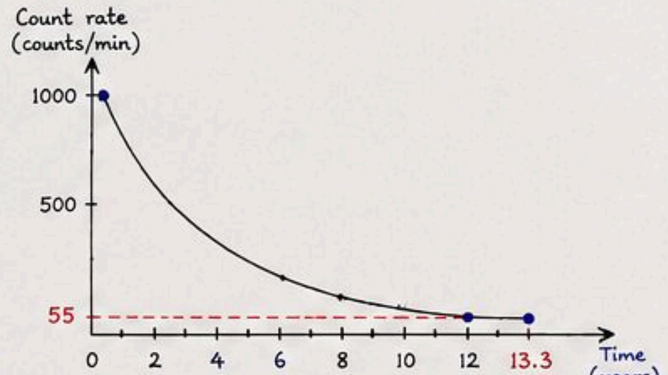
$\frac{1}{100} = \left(\frac{1}{2}\right)^n$

$n = \frac{\log 100}{\log 2} = 6.64$ half-lives

Each half-life = 2 years.

$t = n \times T_{1/2} = 6.64 \times 2 = 13.28$ years

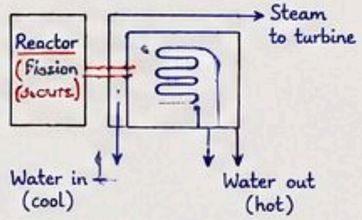
∴ Time ≈ **13.3 years.**



(ii) Addressing the concerns of the surrounding community.

* How water helps in the reactor.

The fission reaction produces a lot of heat. Water absorbs this heat and is boiled into steam. The steam rotates turbines connected to generators to produce electricity. Water also helps to control the heat and prevent the reactor from overheating.



* Economic effects.

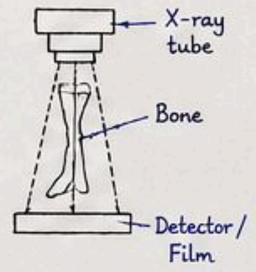
- Reliable electricity supply, job creation, industrial development, improved infrastructure, support for businesses and reduction of dependence on some fossil fuels.

* Health and environmental hazards.

- Radiation can damage cells causing burns, mutations and cancer. Radioactive materials can contaminate air, water and soil if not properly handled. Strict safety measures, shielding and proper disposal of radioactive waste are therefore necessary.

(b) How the device used to locate the fracture works.

- The device is an X-ray machine. X-rays are passed through the injured body part. Bones absorb more X-rays than soft tissues. A detector records the transmitted X-rays and produces an image. The fracture appears on the image because the bone is darker/whiter than the surrounding soft tissue.



3. ITEM 2 [UMTA] – Radioactive tracer for underground water leakage

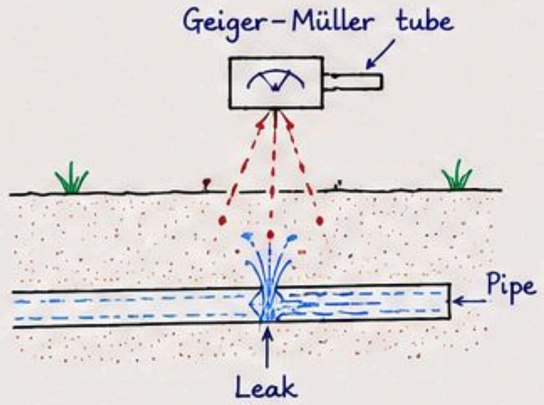
(a) Is the alpha particle method reliable?

No. Alpha particles have very low penetrating power. They can be stopped by a few centimetres of air, clothing or soil. They cannot travel through underground pipes and soil to be detected at the surface. A more penetrating radiation such as gamma rays is more suitable for tracing underground leaks.

Penetrating power
α – very low
β = moderate
γ – very high

(b) (i) How the tracer is used.

A suitable radioactive tracer is introduced into the water in the underground pipe. As the water leaks out at a point, the tracer also leaks out. A Geiger-Müller tube (detector) is moved over the ground, Where a high count rate is obtained, that is the leakage point. This avoids digging many pits.



(ii) How long before the water is approximately safe?

Initial activity, A_o = 1500 counts/min
Background radiation, A_B = 50 counts/min
Safe level ≈ background = 50 counts/min

Half-life, $T_{1/2}$ = 10 minutes

Using $A = A_o \left(\frac{1}{2}\right)^n$

$50 = 1500 \left(\frac{1}{2}\right)^n$

$\frac{1}{30} = \left(\frac{1}{2}\right)^n$

$\log \frac{1}{30} = n \log \frac{1}{2}$

$n = \frac{\log 30}{\log 2} = 4.91$ half-lives

Time, $t = n \times T_{1/2}$

$= 4.91 \times 10$ minutes

$= 49.1$ minutes

$t \approx 49$ minutes

(iii) Safety measures.

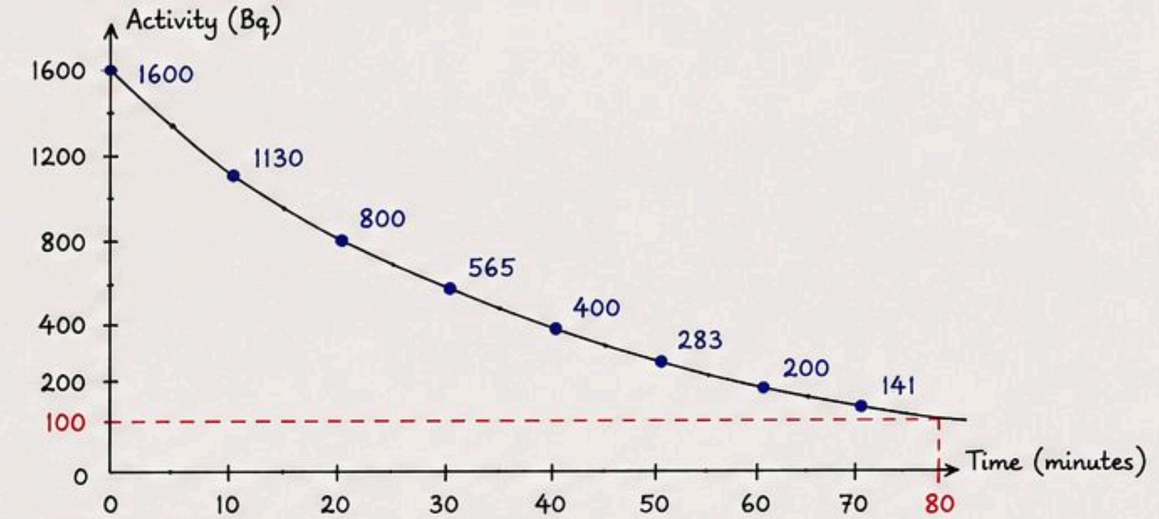
- Use minimum amount of tracer.
- Use tracer with short half-life.
- Keep unnecessary people away.
- Handle using tongs/gloves.
- Use shielding where necessary.
- Monitor using GM tube.
- Prevent contamination of water and food.
- Store in labelled containers.
- Dispose of radioactive waste safely.
- Follow radiation safety rules.

4. ITEM 2 [KAMSSA] – Radioactive contamination of vegetables

(a) Radiations that may have affected the vegetables.

The possible radiations are: alpha (α), beta (β) and gamma (γ).
The exact type cannot be determined from the given data alone.

(b) Graph of activity against time and time taken for vegetables to be safe.



From the graph (or data):
Half-life = time for activity to reduce to half.
1600 → 800 takes 20 minutes ⇒ $T_{1/2}$ = 20 minutes.
To reach 100 Bq: 1600 → 800 → 400 → 200 → 100
This is 4 half-lives.
Time = $4 \times T_{1/2} = 4 \times 20 = 80$ minutes.

Therefore, vegetables become safe after 80 minutes.

(c) Dangers and precautions.

Dangers of exposure:

- Ionising radiation damages body cells.
- It can damage DNA.
- It can cause mutations.
- It increases the risk of cancer.
- It can cause tissue damage and sickness at high doses.

Precautions:

- Do not eat contaminated vegetables.
- Keep away from contaminated areas.
- Prevent radioactive materials entering water sources.
- Use shielding (concrete, lead, etc).
- Minimize time near sources.
- Maximize distance from sources.
- Monitor using radiation detectors.
- Dispose of contaminated materials safely.

Remember: TIME ↓, DISTANCE ↑, SHIELDING ↑

5. ITEM 2 [UTEC] – Radioactive swamp contamination

(a) (i) Estimated time the community should stay away.

Given data:
Initial count rate, A_0 = 625 counts/hour
Required count rate, A = 220 counts/hour
From the table, at 0 h → 625 counts/h
at 90 h → 300 counts/h
at 118 h → 250 counts/h

Assume radioactive decay: $A = A_0 \left(\frac{1}{2}\right)^n$
where n = number of half-lives.
From the table, 625 → 300 in 90 h.
This is close to one half-life.
∴ Half-life, $T_{1/2} \approx 90$ hours
To reach 220 counts/hour:

$$220 = 625 \left(\frac{1}{2}\right)^n$$

$$\left(\frac{1}{2}\right)^n = \frac{220}{625} = 0.352$$

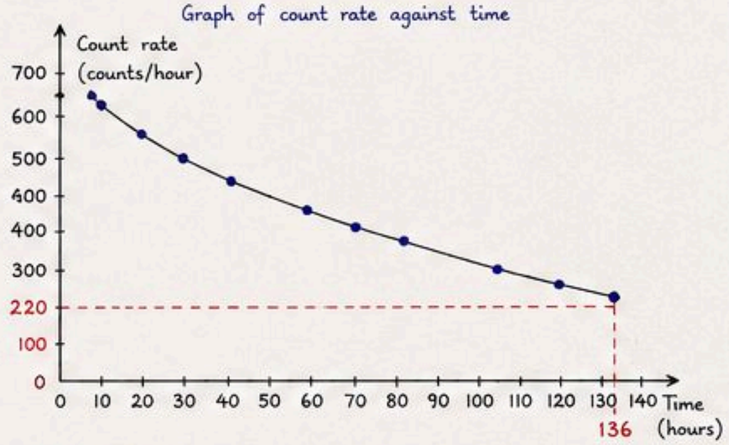
$$n = \frac{\log 0.352}{\log \frac{1}{2}} = 1.51 \text{ half-lives}$$

$$\text{Time} = n \times T_{1/2} = 1.51 \times 90$$
$$= 135.9 \text{ hours} \approx 136 \text{ hours}$$

Time ≈ 136 hours (about 5.7 days)

Time (hours)	0	10	30	46	60	90	118
Count rate (counts/hour)	625	575	500	450	400	300	250

Graph of count rate against time



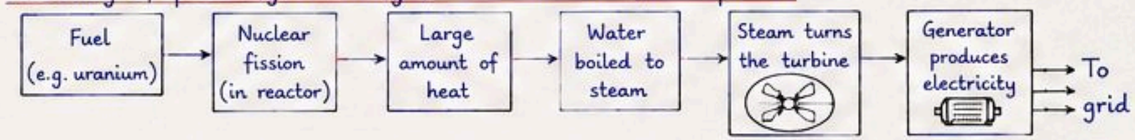
(ii) Will they undergo medical tests?

The half-life is approximately 90 hours
$= \frac{90}{24} = 3.75$ days.
Since this is less than 4 days, they will not necessarily undergo medical tests.

(b) Why should the community stay away?

- The radioactive material emits ionising radiation.
- High radiation can damage body cells, cause mutations and cancer.
- With time, the activity decreases due to radioactive decay.
- Staying away reduces the risk of exposure to harmful radiation.

(c) Possibility of producing electricity from the nuclear fission process.



Energy conversion: Nuclear energy → Heat energy → Kinetic energy → Electrical energy.

(d) Balanced nuclear equation of fission.

$$^{235}_{92}U + {}^{1}_{0}n \longrightarrow {}^{136}_{54}Xe + {}^{97}_{38}Sr + 3\,{}^{1}_{0}n + \text{energy}$$

x = 136 y = 97

(e) Uses and international regulations.

Uses:
- Medical diagnosis (e.g. X-rays)
- Cancer treatment (radiotherapy)
- Food preservation.
- Industrial testing (thickness, leak detection)
- Generation of electricity

International regulations:
- Use under strict licence and control.
- Proper storage, transport and disposal of radioactive materials.
- Monitoring of radiation to keep exposure low.
- International bodies like IAEA ensure peaceful use and prevent misuse of nuclear materials.

6. ITEM 2 [WENSSEC] – Radioactive material in a laboratory

A radioactive substance with activity 750 counts/min spilled in the laboratory. Workers were told to evacuate. An X-ray screening system was installed at the entrance.

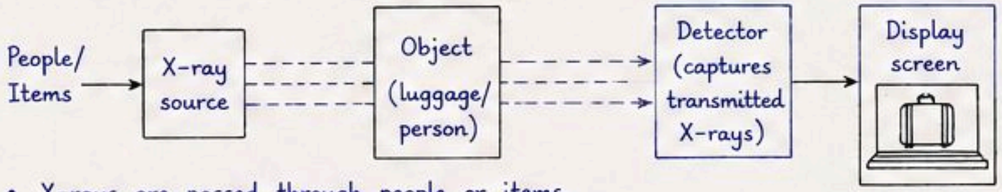
(a) Why were the workers instructed to leave?

- The spilled substance is a source of ionising radiation.
- Ionising radiation can penetrate the human body, damage cells and cause diseases such as cancer.
- Staying in the area increases the risk of exposure.
- Therefore, workers were evacuated to protect their health while experts control and clean the area.

(b) How long will the laboratory remain closed before it is safe to re-enter?

- The time depends on how fast the radioactive material decays.
  Activity decreases with time according to: $A = A_0 \left(\frac{1}{2}\right)^n$
  where A_0 = 750 counts/min, n = number of half-lives.
  Scientists monitor the area using a Geiger-Müller tube.
  The laboratory can be reopened only when the count rate falls to the safe level (near background level).
  (Exact time cannot be given without knowing the half-life and safe level.)

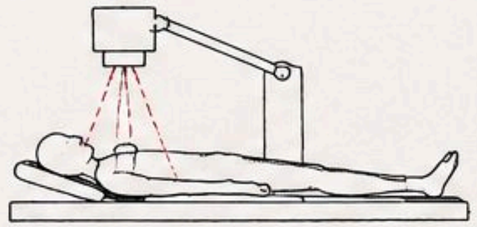
(c) How does the X-ray screening system improve laboratory security?



- X-rays are passed through people or items.
* Different materials absorb different amounts of X-rays.
* The detector converts transmitted X-rays into an image on the screen.
* Items that are dangerous or prohibited can be easily detected.
* This helps to prevent smuggling of weapons, drugs or radioactive materials.

(d) One benefit of radioactive material to the community.

Radioactive materials are used in medicine to treat cancer (radiotherapy). They destroy cancer cells and prevent them from multiplying, helping patients to recover.



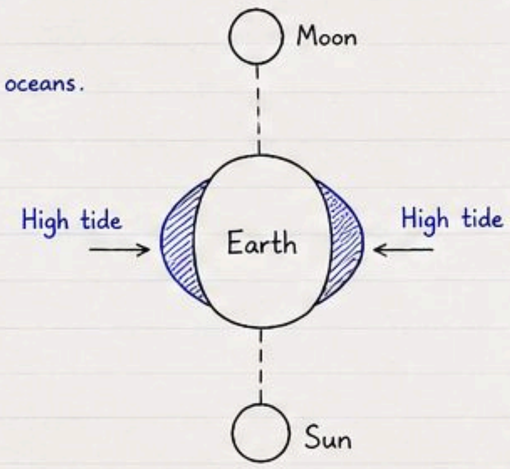
Radioactive source directed at tumour. Cancer cells are destroyed while surrounding healthy tissues are protected.

ITEM 3 [WAKISSHA]

(a)(ii) Why was the water level rising beyond normal?

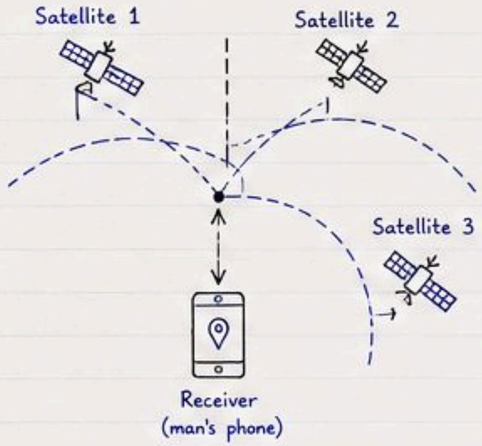
The rise was due to TIDES.

- The Moon's gravitational pull attracts the Earth's oceans.
- The Sun also exerts gravitational attraction.
- This produces periodic high and low tides.
- When the gravitational effects produce a stronger tidal bulge, the sea level rises above its normal level.



(a)(ii) Working of GPS.

1. GPS satellites continuously transmit radio signals containing their position and accurate time.
2. The receiver in the man's phone receives signals from several satellites.
3. It determines the distance from the satellites using the time taken for the signals to arrive.
4. Using TRILATERATION, the receiver calculates its position.
5. The position is displayed on the map, allowing him to locate the mall and find his way back.

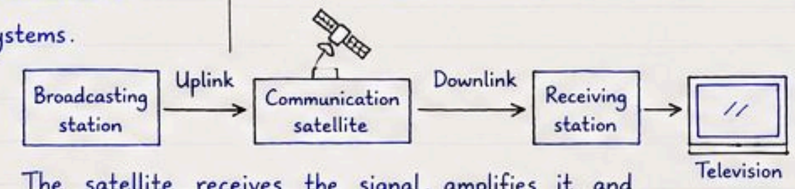


(b) Benefits of rising water levels/tides.

- Helps ships and boats to enter and leave harbours.
- Supports fishing activities.
- Can be used to generate tidal electricity.
- Helps mix nutrients in coastal waters.
- Supports some coastal ecosystems.

(c) How live television broadcasts can cross countries.

A television programme is converted into electrical/digital signals and transmitted using electromagnetic waves.

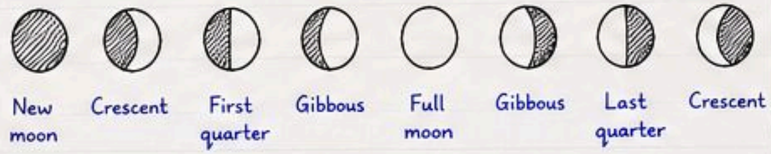


The satellite receives the signal, amplifies it and retransmits it toward the receiving region. This allows programmes from Uganda to be received in distant countries.

ITEM 3 [KAMSSA]

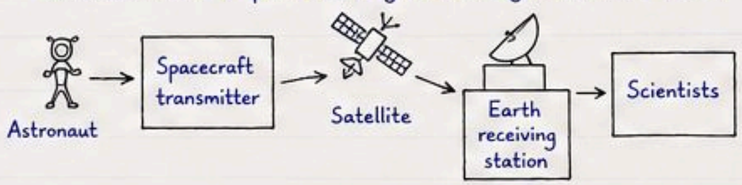
(a) Why does the Moon appear to have different shapes?

The Moon does not actually change shape.
It reflects sunlight. As the Moon revolves around the Earth, the relative positions of the Sun, Earth and Moon change. Therefore, different portions of the Moon's sunlit half are visible from Earth. These different appearances are called PHASES OF THE MOON.



(b)(ii) How can astronauts communicate with Earth?

Communication is possible using electromagnetic radio waves.



Radio waves can travel through the vacuum of space because they do not require a material medium.
The satellite acts as a RELAY STATION, receiving signals and retransmitting them.

(ii) Two advantages of satellites in communication.

1. They can transmit signals over very long distances.
2. They provide communication between places that are difficult to connect directly using cables.

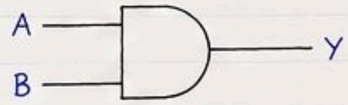
Other advantages:

- Useful for television broadcasting.
- Useful for weather monitoring.
- Useful for navigation.
- Can cover large geographical areas.

(c) Logic gate.

Given: $Y = A \cdot B$

The symbol is an AND gate.



Truth table.

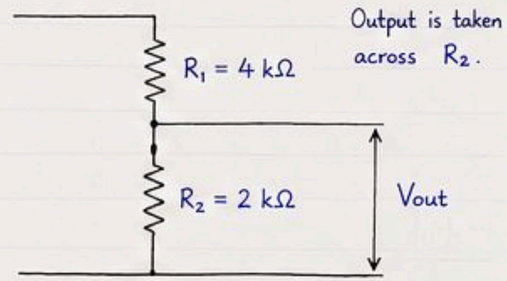
A	B	Y = A · B
0	0	0
0	1	0
1	0	0
1	1	1

Therefore, the emergency transmitter operates only when A = 1 and B = 1.

(d) Determine whether the potential divider works.

Given:

$V_{in} = 12$ V



Potential divider formula: $V_{out} = \frac{R_2}{R_1 + R_2} \times V_{in}$

$V_{out} = \frac{2}{4+2} \times 12$

$V_{out} = \frac{2}{6} \times 12 = 4$ V

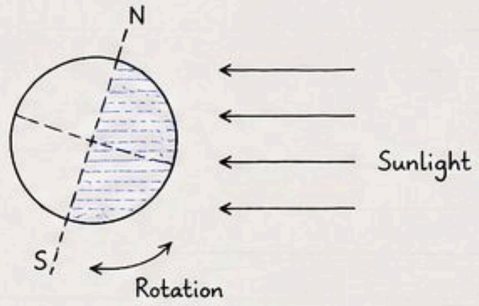
Required: $V_{out} > 3$ V

Since $4 \text{ V} > 3 \text{ V}$

YES, the potential divider will work.

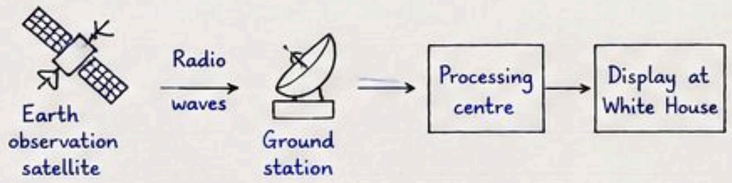
ITEM 3 [UTEC]

(a) Cause of the difference in recorded times.
The times recorded in Iran and the USA are different because of TIME ZONES.
The Earth is divided into 24 time zones.
Different places experience local time at different times because the Earth rotates from West to East.



(b) How could the White House observe the attack?
They could use EARTH OBSERVATION SATELLITES.

1. Satellites orbit the Earth.
2. They carry cameras and sensors to observe the Earth.
3. The satellites collect images/information of events on Earth.
4. The information is sent by radio waves to ground stations.
5. The ground stations receive, process and display the information to users such as the White House.



(c) Distance between Iran and Israel.

Mach number = 13

Speed of sound = 330 m s^{-1}

Speed of missile = Mach no. × speed of sound

$\vec{v}$ = 13 × 330 = 4290 m s^{-1}

Time = 9 min = 9 × 60 = 540 s

Distance, d = v × t

d = 4290 × 540

d = 2,316,600 m

d = 2,316.6 km ≈ 2317 km

Distance ≈ 2317 km

(d) Other components of the Solar System.

- The Sun
- Eight planets
- Dwarf planets
- Natural satellites (moons)
- Asteroids
- Comets
- Meteoroids
- Interplanetary dust and gas

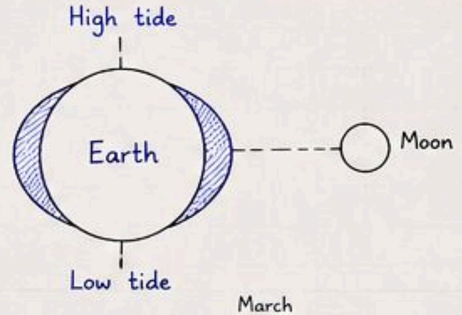
The eight planets are:
Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune.

(e) Other applications of satellites.

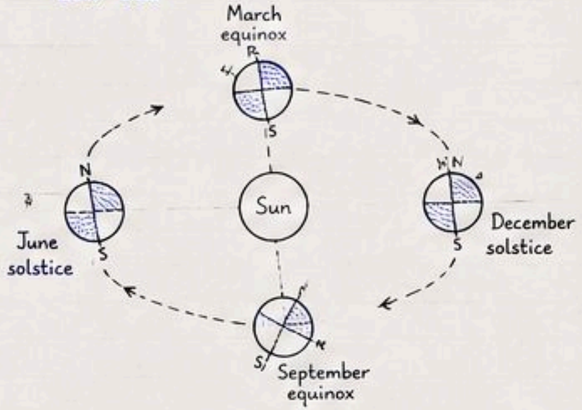
- GPS and navigation
- Weather forecasting
- Television broadcasting
- Telephone communication
- Internet communication
- Mapping and surveying
- Disaster monitoring
- Environmental monitoring
- Military observation
- Scientific research

ITEM 3 [UMTA]

(a) (i) Why do water levels change?
Water levels change mainly due to TIDES.
The Moon's gravitational pull attracts the sea water. The Sun also attracts it but to a smaller extent. This causes water to rise (high tide) and fall (low tide) periodically.



(ii) Why does Canada have longer days in July?
The Earth's axis is tilted at about 23½°.
As the Earth revolves around the Sun, the tilted axis causes different hemispheres to receive different amounts of sunlight at different times of the year.
In July, the Northern Hemisphere (where Canada is located) is tilted towards the Sun. Therefore, Canada receives more direct sunlight and has longer days and shorter nights.

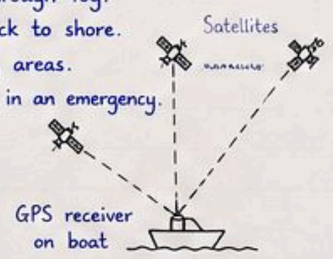


(iii) How does the Sun produce enormous energy?
The Sun produces energy by NUCLEAR FUSION.
In its core, hydrogen nuclei combine under extremely high temperature and pressure to form helium nuclei.
A small amount of mass is converted into a large amount of energy according to $E = mc^2$. This energy is released as heat and light.

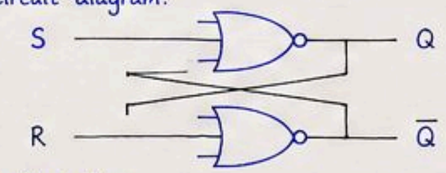
*Do stars become hotter or cooler as they grow older?
Stars pass through different stages in their life cycle. Their temperature does not simply increase or decrease continuously. A star's temperature depends on its mass and the stage it is in. For example, it may become much hotter in some stages and later expand into a cooler giant-star.

(b) How can fishermen use GPS?
A GPS receiver on the boat receives signals from several satellites. It calculates the boat's position (latitude and longitude) and shows it on a map.
Fishermen can therefore:

- Know their exact location.
- Navigate safely through fog.
- Find their way back to shore.
- Mark good fishing areas.
- Locate another boat in an emergency.



(c) Bistable switch using two NOR gates.
Circuit diagram.



Truth table.

S	R	Q(n+1)	$\overline{Q}$(n+1)	State
0	0	Q(n)	$\overline{Q}$(n)	No change
0	1	0	1	Reset
1	0	1	0	Set
1	1	0	0	Invalid

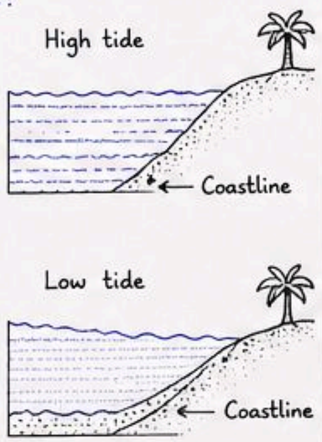
Note: S = R = 1 is the forbidden (invalid) state.

ITEM 3 [WENSSEC]

(a) Why did the coastline appear to shift?

The coastline appeared to shift because of TIDES.

- At high tide, sea water moves towards the land making the coastline appear nearer.
- At low tide, water moves away from the land making the coastline appear farther.



Benefits of tides

- Fishing
- Helps in navigation of ships
- Harbour operations
- Generation of tidal electricity
- Helps in mixing nutrients in coastal waters

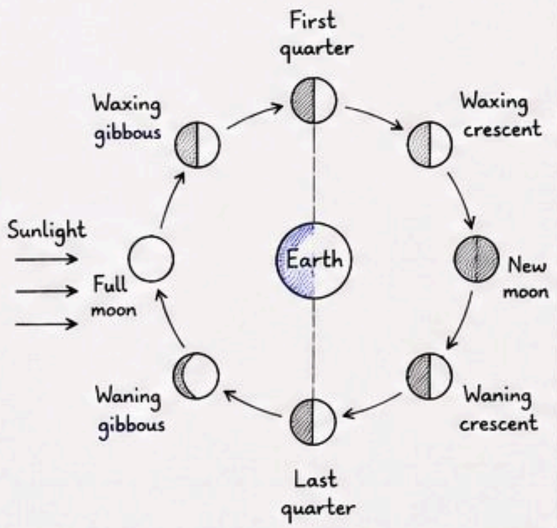
(b) How do stars come into existence?

Stars form from huge clouds of gas and dust called nebulae.

1. Gas and dust come together by gravity.
2. They become denser.
3. Temperature and pressure increase.
4. A protostar forms.
5. Nuclear fusion begins in the core.
6. A star is formed.

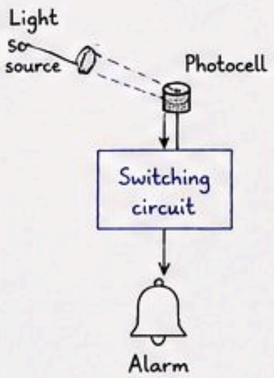
(c) Why does the Moon's appearance change?

The Moon reflects sunlight. As it revolves around the Earth, different portions of its illuminated half become visible from Earth. These different appearances are called PHASES OF THE MOON.



(d) How does a photocell alarm work?

A photocell detects changes in light intensity.



(e) How did the relatives receive the telephone call?

The telephone converts the caller's voice into electrical signals. These signals are transmitted through communication networks to the receiver.

Caller's phone → Mobile network / exchange → Switching centre / network → Telco in the other country → Relative's phone

Communication may use fibre-optic cables, microwave links or communication satellites to connect distant places.

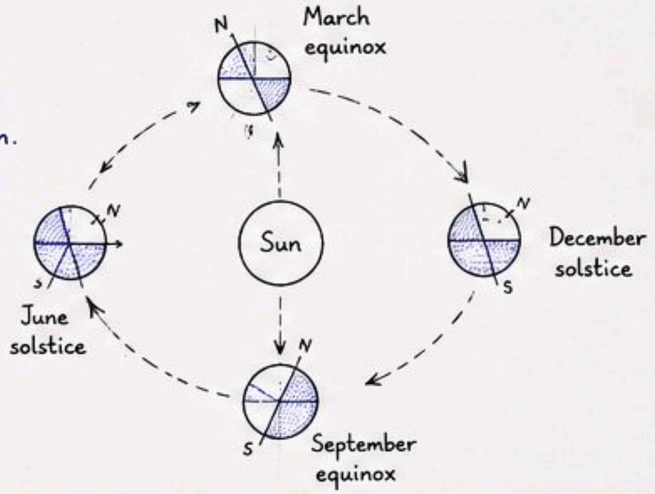
ITEM 3 [MASAKA]

(a) Cause of different seasons.

Seasons are caused by:

1. The tilt of the Earth's axis (about 23.5°).
2. The revolution of the Earth around the Sun.

When a hemisphere is tilted towards the Sun, it receives more direct sunlight, has longer days and experiences summer.
When tilted away, it receives less direct sunlight, has shorter days and experiences winter.



(b) Greenhouse automatic control system.

Conditions:

A = 1 when temperature > 30°C
B = 1 when humidity < 30%

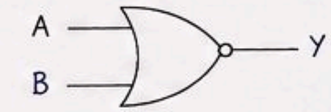
The fan and sprinkler should operate if either condition occurs.
This is an OR operation.

$$Y = A + B$$

Truth table

A	B	Y = A + B	(Fan/Sprinkler)
0	0	0	Off
0	1	1	On
1	0	1	On
1	1	1	On

OR gate symbol



Other applications

- Automatic fire alarms
- Automatic street lights
- Security systems
- Automatic irrigation systems
- Refrigerator temperature control
- Industrial temperature control

(c) Why choose LED bulbs?

LED bulbs are preferred because they:

- Use less electrical energy.
- Produce less unwanted heat.
- Have a long operating life.
- Are more efficient than filament bulbs.
- Convert a greater proportion of electrical energy into light.
- Reduce electricity costs.

	LED bulb	Filament bulb
Energy use	Low	High
Heat produced	Low	High
Life span	Long	Short
Efficiency	High	Low

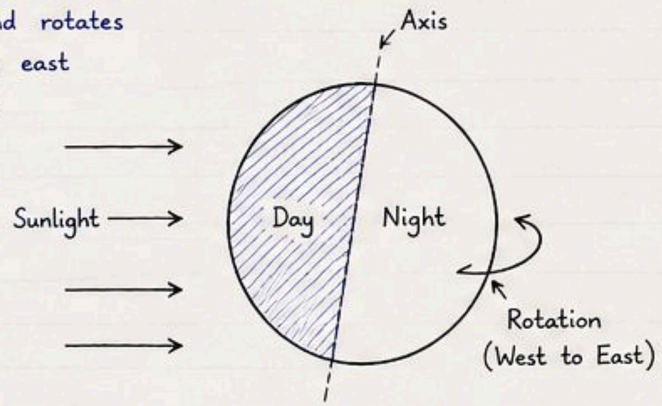

LED bulb


Filament bulb

ITEM 3 [JJEB]

(a) (i) Why does Earth have one side light and another dark?

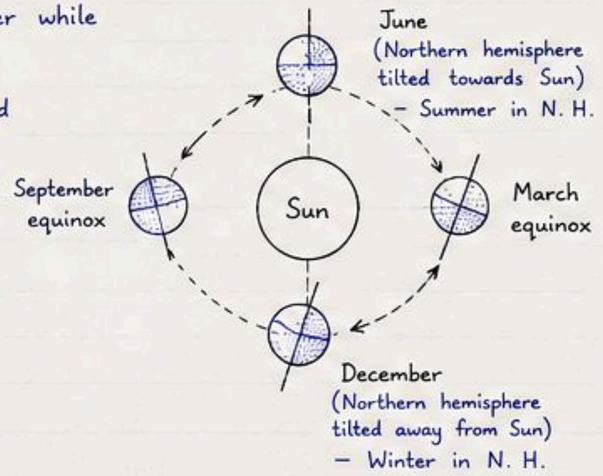
Earth is nearly spherical and rotates on its axis from west to east once every about 24 hours.
The side facing the Sun receives sunlight and has day.
The side facing away from the Sun receives little or no direct sunlight and has night.
Therefore we have day and night.



One complete rotation = about 24 hours.

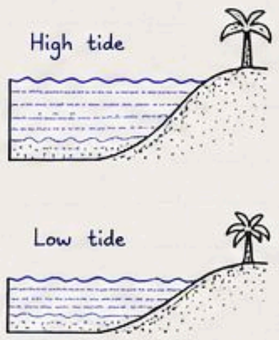
(iii) Why can the tourist experience summer while another country experiences winter?

This is because the Earth's axis is tilted at about 23½° and the Earth revolves around the Sun.
When a hemisphere is tilted towards the Sun, it receives more direct sunlight, has longer days and therefore experiences summer. The opposite hemisphere is tilted away from the Sun and has less direct sunlight, shorter days and experiences winter.



(iiii) Why is the ocean low in the morning and high in the evening?

This is due to TIDES.
Tides are caused mainly by the gravitational pull of the Moon and to a lesser extent, the Sun on the Earth's oceans.
As Earth rotates, a place on Earth passes through two high tides and two low tides each day.

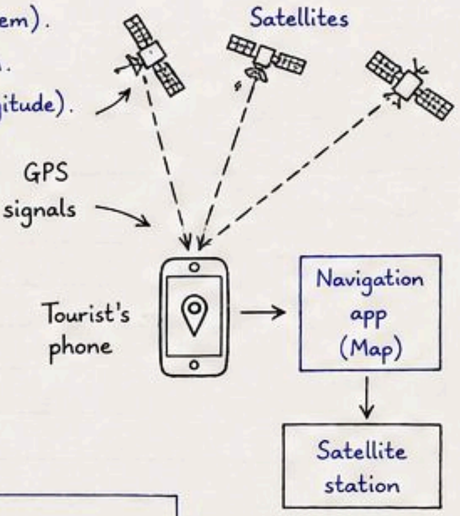


(b) How can the navigation app help the tourist reach the satellite station?

The navigation app uses GPS (Global Positioning System).
It receives signals from several satellites orbiting Earth.
The phone calculates the tourist's position (latitude and longitude).
The app then:

1. Identifies the tourist's current position.
2. Identifies the location of the satellite station.
3. Calculates the best route.
4. Displays the route or directions on the map.
5. Guides the tourist using voice or map instructions.

This helps the tourist reach the station.

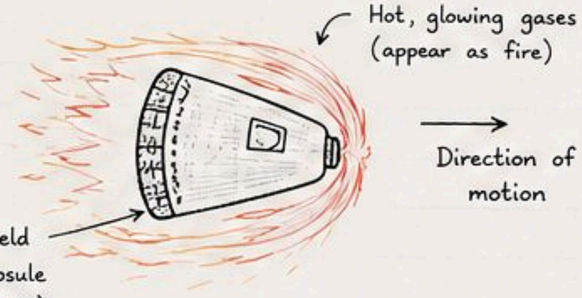


Why should Uganda invest in space exploration?

- It improves communication (satellites for TV, internet and phone).
- It improves weather forecasting and disaster monitoring.
- It helps in navigation and mapping.
- It helps in environmental and resource monitoring.
- It promotes science, technology and innovation.
- It develops skills and creates jobs.
- It increases national security and independence.
- It inspires education and research.

(c) Why does the returning capsule appear to emit fire?

When the capsule enters the Earth's atmosphere, it moves at a very high speed.
The air molecules in front of the capsule are compressed and collide with it.
This creates a lot of friction and heat.
The outside of the capsule (heat shield) becomes extremely hot and glows.
The surrounding hot gases also glow.
This glow looks like fire.
The heat shield melts or burns away slowly but protects the capsule and astronauts from the extreme heat.

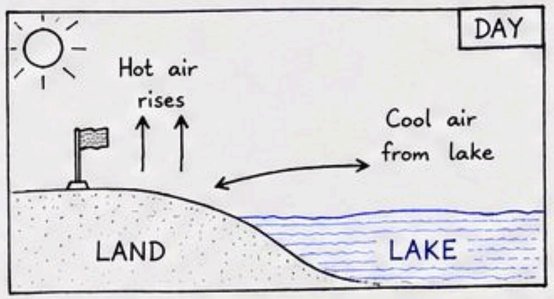


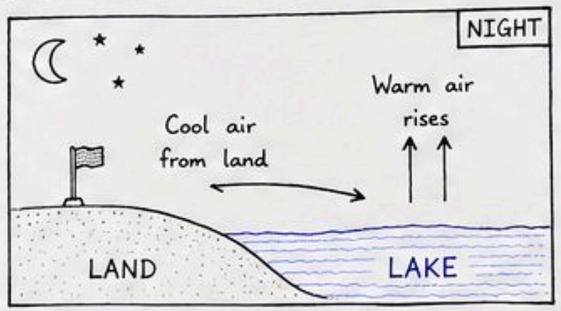
PAIR 1 – WAKISSHA

ITEM 4

(a) Why did the flag fly towards land during the day and towards the sea at night?

This is due to land and sea breezes.





During the day (sea breeze)

- Land heats up faster than water.
- Air above the land becomes hot and rises.
- Cooler, denser air from the lake moves towards the land.
- This pushes the flag towards land.

At night (land breeze)

- Land cools faster than water.
- Air above the lake becomes warm and rises.
- Cooler air from the land moves towards the lake.
- This pushes the flag towards the sea.

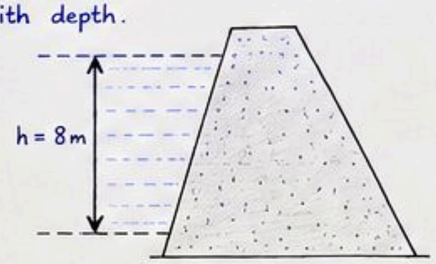
(b) Why are dam walls thicker at the bottom? Advise him on filling the dam.

Liquid pressure increases with depth.

$$P = \rho g h$$

At depth $h = 8$ m,

$P = 1000 \times 10 \times 8 = 80{,}000$ Pa



Maximum pressure supported = 70,000 Pa

Since 80,000 Pa > 70,000 Pa, the dam should not be filled to 8 m.

Maximum safe depth,

$$h = \frac{P}{\rho g} = \frac{70{,}000}{1000 \times 10}$$

$h = 7$ m

Therefore, the dam should be filled to a depth not more than 7 m.

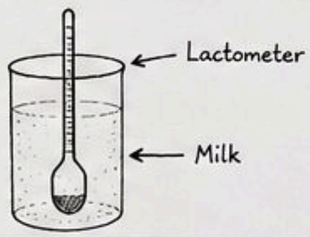
(c) How can he test whether the milk has been diluted?

Use a lactometer (or hydrometer).

It measures the density (specific gravity) of milk.

If water is added, the density decreases.

Compare the reading with the normal value for genuine milk.



(d) Applications of atmospheric pressure.

- Drinking through a straw
- Syringes
- Droppers
- Suction pumps
- Vacuum cleaners
- Siphons
- Barometers
- Force pumps

ITEM 5

(a) Why is it hotter at the bottom of the mountain?

Temperature generally decreases with increase in altitude. Air pressure is lower at high altitude, so rising air expands and cools. Therefore the foot of the mountain is generally warmer than the top.



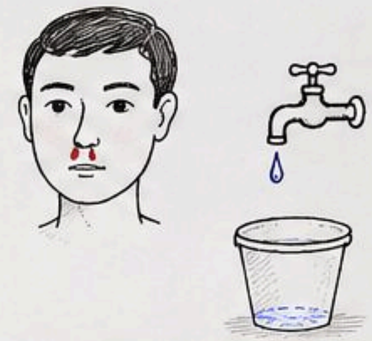
(b) How can the low atmospheric pressure at high altitude contribute to nose bleeding and water shortage?

Nose bleeding:

Lower pressure makes the pressure difference across delicate blood vessels greater, which may cause them to burst easily.

Water shortage:

Water sources are limited and evaporation/drainage is faster in mountainous areas.



(c) Determine the height of the mountain.

Given: $\Delta P = 10{,}000$ Pa

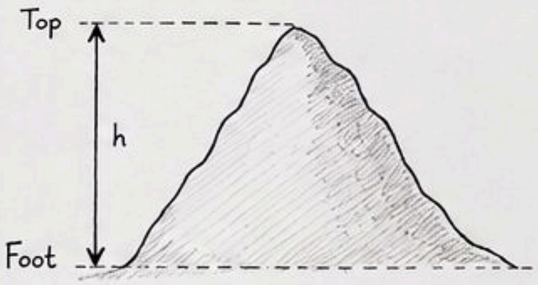
$\rho = 1.25$ kgm^{-3}

$g = 10$ ms^{-2}

Using: $\Delta P = \rho g h$

$$h = \frac{\Delta P}{\rho g} = \frac{10{,}000}{1.25 \times 10}$$

$h = 800$ m



(d) Calculate the power required by the pump.

Flow rate = 300 L/min = 0.3 m^3/min

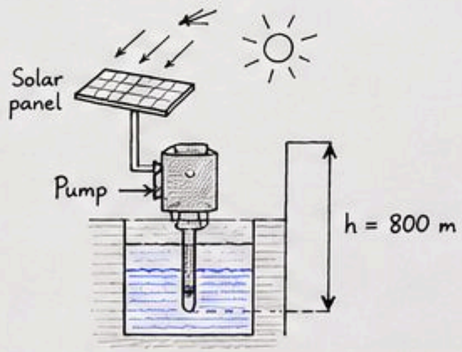
$$Q = \frac{0.3}{60} = 0.005 \text{ m}^3/\text{s}$$

Mass flow rate, $\dot{m} = \rho Q = 1000 \times 0.005 = 5$ kg/s

Power, $P = \dot{m} g h$

$= 5 \times 10 \times 800$

$P = 40{,}000$ W $= 40$ kW



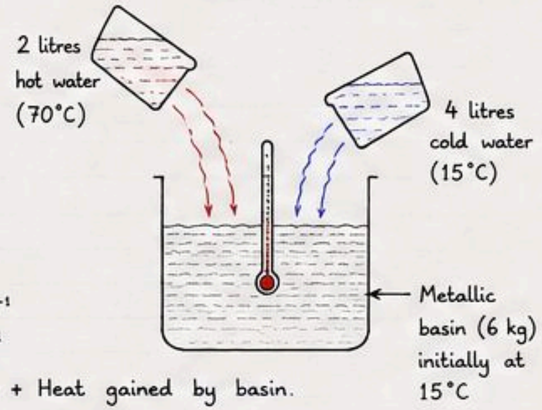
The pump requires a power of approximately 40 kW (assuming 100% efficiency).

PAIR 3 – KAMSSA

ITEM 4

(a) Determine whether the final temperature of the water mixture is suitable for the baby.

Let final temperature of mixture = T °C



Given :

Mass of hot water, m_h = 2 litres = 2 kg
Temperature of hot water, = 70 °C
Mass of cold water, m_c = 4 litres = 4 kg
Temperature of cold water, = 15 °C
Mass of basin, m_b = 6 kg
Initial temperature of basin, = 15 °C
Specific heat capacity of water, c_w = 4200 $J\,kg^{-1}\,K^{-1}$
Specific heat capacity of basin, c_b = 900 $J\,kg^{-1}\,K^{-1}$

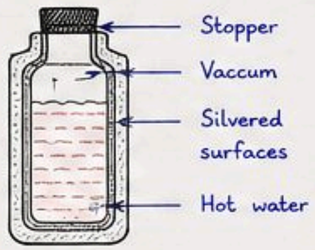
Heat lost by hot water = Heat gained by cold water + Heat gained by basin.

$$m_h\,c_w\,(70 - T) = m_h\,c_w\,(T - 15) + m_b\,c_b\,(T - 15)$$
$$2\,(4200)\,(70 - T) = 4\,(4200)\,(T - 15) + 6\,(900)\,(T - 15)$$
$$84\,000 - 8400\,T = 16\,800\,(T - 15) + 5400\,(T - 15)$$
$$84\,000 - 8400\,T = 22\,200\,T - 332\,100$$
$$84\,000 + 332\,100 = 22\,200\,T + 8400\,T$$
$$416\,100 = 30\,600\,T$$
$$T = 13.6\,°C$$
$$T = 30.1\,°C$$

Recommended temperature for baby = 29 °C – 35 °C

The final temperature (≈ 30.1 °C) is suitable for the baby.

(b) Advise the family on ways of keeping the hot water warm while they are away welcoming the visitor.

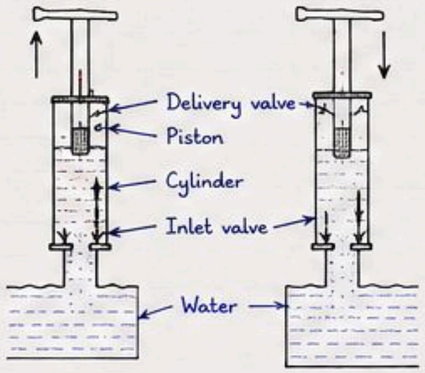


- Put the hot water in a vacuum flask.
- Cover the container with a lid.
- Wrap the container with insulating material (e.g. cloth, foam).
- Keep the container in a warm place.
- Reduce the exposed surface area.
- Do not stir the water unnecessarily.

(c) Explain how a force pump works to raise water from the well and deliver it to a higher level.

When the handle is pulled up, the piston moves up and the volume in the cylinder increases. Pressure in the cylinder falls below atmospheric pressure, so the inlet valve opens and water is pushed into the cylinder by atmospheric pressure.

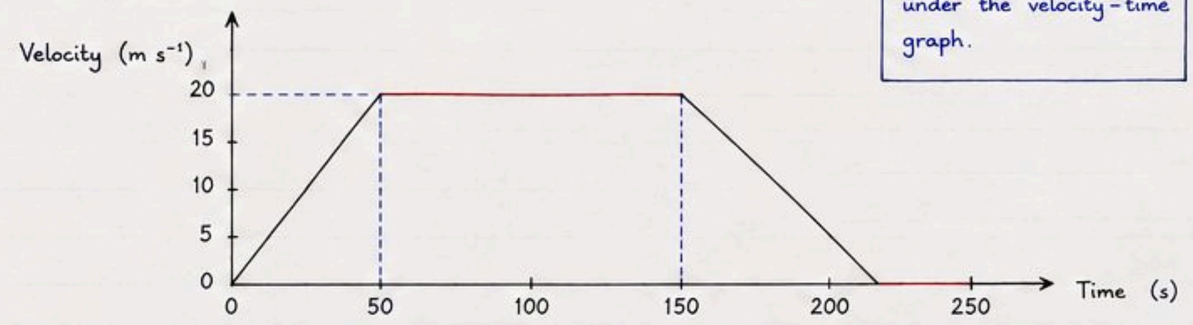
When the handle is pushed down, the piston moves down and the volume decreases. Pressure in the cylinder increases, the inlet valve closes and the delivery valve opens. Water is forced out through the delivery pipe to a higher level.



ITEM 5

(a) Using the velocity-time graph below, determine the average speed of the vehicle for the whole journey and state whether the driver was penalized for exceeding 10 m s^{-1}.

Distance travelled = area under the velocity-time graph.



Area 1 (triangle) = $\frac{1}{2}$ × 50 × 15 = 375 m
Area 2 (rectangle) = 100 × 15 = 1500 m
Area 3 (triangle) = $\frac{1}{2}$ × 50 × 15 = 375 m
Total distance = 375 + 1500 + 375 = 2250 m
Total time = 200 s

$$\text{Average speed} = \frac{\text{total distance}}{\text{total time}} = \frac{2250}{200} = 11.25\ m\,s^{-1}$$

Since 11.25 m s^{-1} > 10 m s^{-1}, the driver was penalized for exceeding the speed limit.

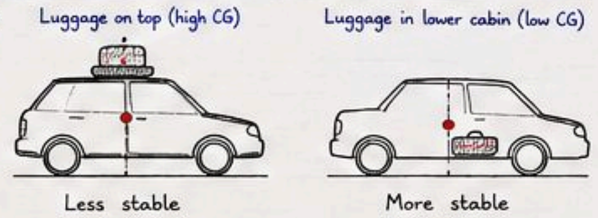
(b) Explain the importance of seat belts to passengers in a moving vehicle.

- When a vehicle stops suddenly, a passenger's body tends to continue moving forward due to inertia.
- The seat belt applies a restraining force that stops the passenger from hitting the dashboard, windscreen or other hard parts.
- It keeps the passenger in the seat and reduces the risk of serious injury or death.
- It spreads the force over the stronger parts of the body (chest and hips).



(c) Give one reason why placing luggage in the lower cabin instead of on the top rack is safer.

- It lowers the centre of gravity of the vehicle.
- This makes the vehicle more stable and reduces the chance of overturning, especially when turning or braking.



(d) The candidate's temperature was read as 310.6 K by the nurse. Determine the temperature in °C. Should he pay for the further malaria test if the threshold for the test is 37.5 °C?

Convert Kelvin to Celsius:

$$T(°C) = T(K) - 273$$
$$T(°C) = 310.6 - 273$$
$$T = 37.6\,°C$$

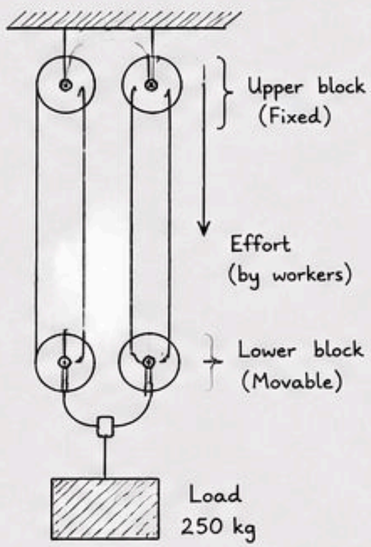
Since 37.6 °C > 37.5 °C (threshold), the candidate is required to pay for the further malaria test.

NOTE: The answers above are based on the velocity – time graph given in the mock paper.

PAIR 2 – JJEB

ITEM 4

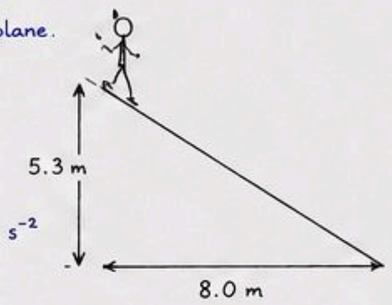
(i) Explain how the pulley system should be arranged and how workers should use the system whose efficiency is not less than 71%.



- Two pulleys are fixed in the upper block and two in the lower (movable) block.
- The rope is fixed at the upper block, passes round the lower pulleys, then up round the upper pulleys and the free end is pulled downwards.
- The lower block (with the load) rises when workers pull the free end.
- To have efficiency ≥ 71%:
  - Use strong, smooth and light rope.
  - Keep pulleys well oiled so they rotate freely.
  - Make sure the pulleys are well aligned.
  - Pull the rope continuously and in the correct direction.
  - Avoid jerking and unnecessary friction.

(ii) Joby monitoring the pulley system should always reach the pulley area at a velocity not less than 4.3 m s^{-1}. Will he manage the task?

To answer, first find the velocity he attains on the inclined plane.



Using the principle of conservation of energy:

$$mgh = \tfrac{1}{2}mv^2$$

$$v = \sqrt{2gh}$$

From the diagram, height of plane, $h = 5.3$ m. $g = 10$ m s^{-2}

$$v = \sqrt{2 \times 10 \times 5.3} = \sqrt{106}$$

$$v = 10.3 \text{ m s}^{-1}$$

Since 10.3 m s^{-1} > 4.3 m s^{-1}, Joby will manage the task.

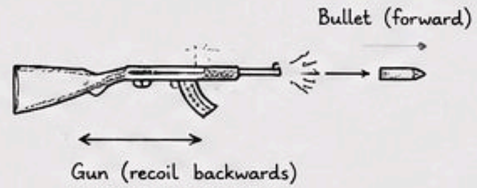
Note: Efficiency of the pulley system = 71% or more.

$$\text{Efficiency} = \left(\frac{\text{Useful work output}}{\text{Work input}}\right) \times 100\% \geq 71\%$$

ITEM 5

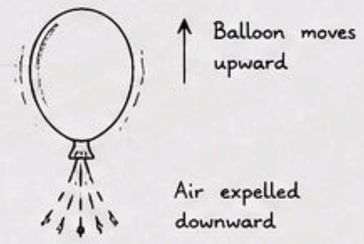
(a) (i) Why does a gun move backwards when fired?

The system (gun + bullet) initially has zero momentum.
When the bullet is fired forward, it gains forward momentum.
To conserve momentum, the gun receives an equal and opposite momentum and moves backwards. This is called recoil.



(ii) Why does the balloon move upward when air is released?

Air is expelled downward from the balloon.
According to Newton's third law, for every action there is an equal and opposite reaction.
The downward pushing of air produces an upward reaction force on the balloon, causing it to move up.



(iii) Did Jalia win?

Height of Jalia = 1.4 m Time to reach maximum height = 1.2 s $g = 10$ m s^{-2}

Speed of ball at maximum height is zero.

Initial speed $u = gt = 10 \times 1.2 = 12$ m s^{-1}

Height gained above hand:

$h = ut - \frac{1}{2}gt^2 = (12 \times 1.2) - \frac{1}{2}(10)(1.2)^2 = 14.4 - 7.2 = 7.2$ m

Total maximum height above the ground = $h + 1.4 = 7.2 + 1.4 = 8.6$ m

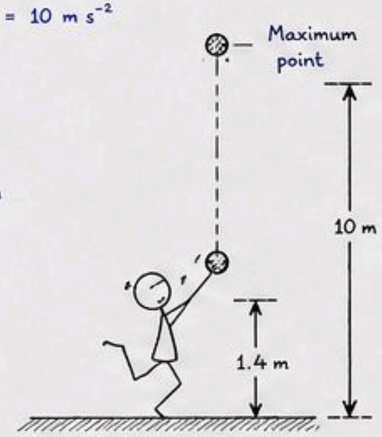
Since 8.6 m < 10 m, Jalia did not win.

Required height above hand to win = $10 - 1.4 = 8.6$ m

Using $v^2 = u^2 - 2gh$, at maximum height $v = 0$.

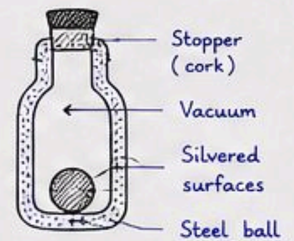
$0 = u^2 - 2(10)(8.6) \Rightarrow u = \sqrt{172} = 13.1$ m s^{-1}

Required initial speed ≈ 13.1 m s^{-1}



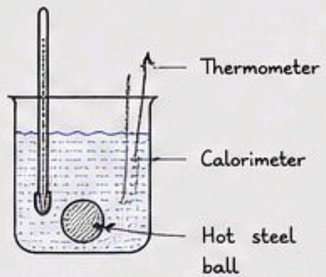
(b) (i) Why can the steel ball remain hot for a long time in the flask?

- The flask has a vacuum between the walls, so conduction and convection are greatly reduced.
- The shiny (silvered) surfaces reduce heat loss by radiation.
- The stopper reduces convection.
- Therefore, very little heat is lost to the surroundings.



(ii) Give one way Jalia can determine the specific heat capacity of the steel ball.

1. Measure the mass of the steel ball (m_b).
2. Heat the ball to a known high temperature (e.g. 100°C).
3. Put it quickly into water of known mass (m_w) and known initial temperature (θ_1) in a well-insulated calorimeter.
4. Stir and record the final equilibrium temperature (θ_f).
5. Calculate the specific heat capacity using: $m_b c_b (\theta_{b,i} - \theta_f) = m_w c_w (\theta_f - \theta_1)$
6. Hence, $c_b = \dfrac{m_w c_w (\theta_f - \theta_1)}{m_b (\theta_{b,i} - \theta_f)}$



PAIR 4 – UMTA

ITEM 4

(i) Assess whether the spring meets safety requirements based on its Young's modulus Hodnlus.

Young's modulus, $E = \frac{\text{stress}}{\text{strain}} = \left(\frac{\frac{F}{A}}{\frac{\Delta L}{L}}\right) = \frac{FL}{A\Delta L}$

Original length, L = 15.0 cm = 0.150 m
Final length at maximum load = 17.5 cm
Extension, ΔL = 17.5 − 15.0 = 2.5 cm = 0.025 m
Cross-sectional area, A = 20 cm^2 = 20 × 10^{-4} m^2 = 0.0020 m^2

From the given load-extension graph,
maximum load, F = 100 N (reading at 2.5 cm extension)

$$E = \frac{FL}{A\Delta L} = \frac{100 \times 0.150}{0.0020 \times 0.025}$$
$$= \frac{15}{0.00005} = 3.0 \times 10^5 \text{ N m}^{-2}$$

Required minimum Young's modulus = 3.11×10^5 N m^{-2}
Since $3.0 \times 10^5 < 3.11 \times 10^5$,

The spring does NOT meet the safety requirement.

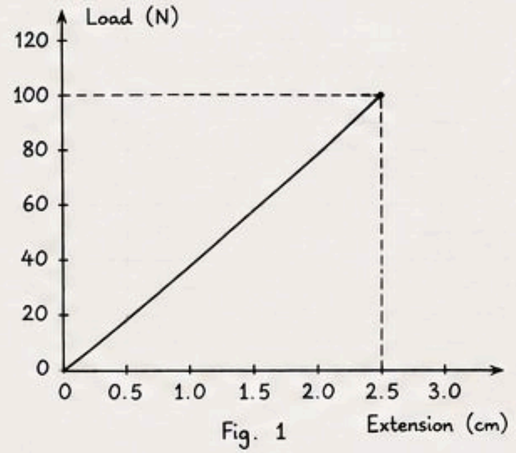


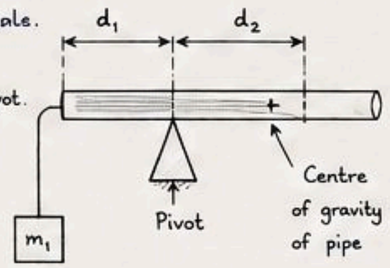
Fig. 1

(iii) Explain how the mass of the hollow pipe can be found using the moriciale.

1. Place the hollow pipe horizontally on a knife-edge (pivot).
2. Suspend a known mass (m_1) at a measured distance (d_1) on one side of the pivot.
3. Move the pipe until it is balanced (horizontal).
4. Measure the distance (d_2) of the pipe's centre of gravity from the pivot.
5. By principle of moments: clockwise moment = anticlockwise moment.

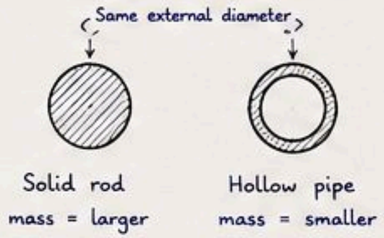
$m_1 g d_1 = m_p g d_2$ (g cancels out)

$\therefore$ mass of pipe, $m_p = \frac{m_1 d_1}{d_2}$



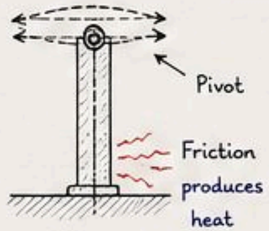
(iiii) Justify the use of a hollow pipe instead of a solid one.

- A hollow pipe has a high strength-to-weight ratio.
- It is lighter, hence easier to handle and reduces load on the structure.
- Material is saved, making it economical.
- It resists bending and torsion well due to the material being away from the centre.
- It can also allow passage of cables, fluids or other components.



(iv) Explain why heat is produced at the pivot.

When the seesaw moves up and down, the surfaces at the pivot rub against each other. Friction opposes motion and converts some of the mechanical energy into thermal energy (heat). This raises the temperature at the pivot. Hence heat is produced.



ITEM 5

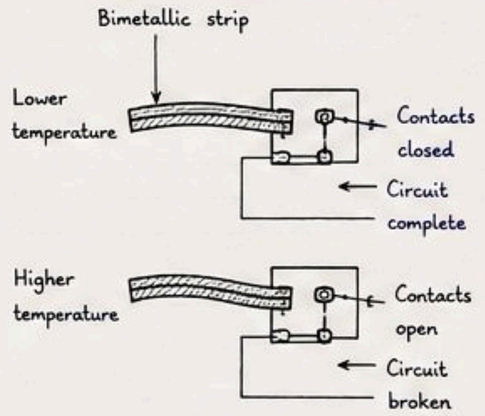
(a) (i) How does a thermostat work? State its applications.

A thermostat contains a bimetallic strip made of two metals with different coefficients of expansion.

- When temperature rises, the strip bends towards the metal with the lower expansion.
- This opens the electrical contacts and the heater switches off.
- When temperature falls, the strip bends back and closes the contacts. The heater switches on again.

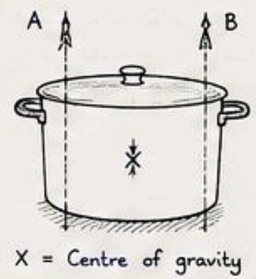
This keeps the temperature nearly constant.

Applications:
Electric iron, refrigerator, electric kettle, oven, geyser, water heater, incubator.



(b) How can you find the centre of gravity of the cooking pot experimentally?

1. Suspend the pot from a point A using a string.
2. Hang a plumb line from the same point and mark the vertical line on the pot.
3. Suspend the pot from another point B.
4. Again hang a plumb line and mark the vertical line.
5. The point where the two lines intersect is the centre of gravity.



(c) Why is it important that the pot has low heat capacity?

- It requires less heat energy to increase its temperature.
- It heats up quickly, saving energy and time.
- It cools down quickly when removed from the heat.
- Therefore cooking is more efficient and economical.

Heat capacity, C = mc
Low C ⇒ less heat needed for a given temperature rise.

(d) (i) Calculate the total heat energy required.

Given:
Mass of sauce, m = 1.0 kg
Initial temperature = −20°C
Final temperature = 90°C
c_{ice} = 2100 J kg^{-1} K^{-1}
L_f = 3.34×10^5 J kg^{-1}
c_{water} = 4200 J kg^{-1} K^{-1}
Mass of pot = 0.20 kg
c_{pot} (aluminium) = 900 J kg^{-1} K^{-1}
Temperature of pot initially = −20°C
Final temperature = 90°C
1 K = 1°C

Stages:
1. Ice −20°C → 0°C
2. Melting at 0°C
3. Water 0°C → 90°C
4. Heating the pot −20°C → 90°C

Stage 1 (ice to 0°C): $Q_1 = mc_{ice}\Delta T = 1 \times 2100 \times (0 - (-20)) = 42\,000$ J
Stage 2 (melting): $Q_2 = mL_f = 1 \times 3.34 \times 10^5 = 334\,000$ J
Stage 3 (water 0°C to 90°C): $Q_3 = mc_w\Delta T = 1 \times 4200 \times (90 - 0) = 378\,000$ J
Stage 4 (heating pot −20°C to 90°C): $Q_4 = mc_p\Delta T = 0.20 \times 900 \times (90 - (-20))$
$= 0.20 \times 900 \times 110 = 19\,800$ J

Total heat energy, $Q = Q_1 + Q_2 + Q_3 + Q_4$
$= 42\,000 + 334\,000 + 378\,000 + 19\,800$
$= 773\,800$ J

(ii) Is the budget sufficient?

Energy supplied by 50 000 J costs Shs. 200

Cost required = $\left(\frac{773\,800}{50\,000}\right) \times 200$
$= 15.476 \times 200$
$= $ Shs. 3 095.20

Budget = Shs. 3 500

Since Shs. 3 095.20 < Shs. 3 500, the budget is sufficient.

PAIR 5 – UTEC

ITEM 4

(a) (i) Explain the physical effect experienced by John when the car started and stopped, and how it could be limited.

The effect is INERTIA.

- When the car starts suddenly, the car moves forward but John's body tends to remain at rest, so he is pushed backward and hits the seat.
- When the car stops suddenly, his body tends to continue moving forward, so he hits the front seat.

When car starts



When car stops



How it can be limited:

- By wearing a seat belt.
- By using head rests.
- By starting and stopping the car gradually.
- By restraining the body with safety devices.

(ii) Give applications of this effect in daily life.

- Seat belts in cars.
- Head rests in vehicles.
- Shaking dust from clothes.
- Beating carpets.
- A hammer driving a nail.
- Passengers lurching forward when a bus stops suddenly.
- Padding in helmets.

(b) Inform the driver whether he was billed for over speeding by the automated machine.

The driver accelerated from rest at $0.1\ m\ s^{-2}$ for 4 minutes.

Time, $t = 4\ min = 240\ s$

Initial velocity, $u = 0$

Acceleration, $a = 0.1\ m\ s^{-2}$

Maximum speed reached,

$v = u + at = 0 + 0.1 \times 240 = 24\ m\ s^{-1}$

Convert to $km\ h^{-1}$:

$v = 24 \times 3.6 = 86.4\ km\ h^{-1}$

Speed limit $= 90\ km\ h^{-1}$

Since $86.4\ km\ h^{-1} < 90\ km\ h^{-1}$, the driver was NOT over speeding and should NOT be billed by the machine.

(c) Help John appreciate the driver's decision regarding the position of the luggage.

- Placing luggage in the lower cabin lowers the centre of gravity.
- This increases the stability of the car.
- It reduces the chance of the car overturning, especially on corners or when braking suddenly.



(d) Determine whether the candidate was required to pay for the further malaria test based on his temperature.

Temperature was read as 310.6 K.

Convert to °C:

$T\ (°C) = T(K) - 273$

$T = 310.6 - 273 = 37.6°C$

If the hospital rule is that the further test is paid for when temperature exceeds 37.5°C (threshold), then

$37.6°C > 37.5°C$ ≤ [illegible]

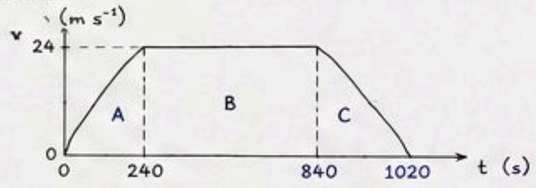
Therefore, he is required to pay for the further malaria test.

(e) With reference to a sketch velocity–time graph, determine if the fare charged by the driver was ideal based on the distance covered.

Acceleration phase: $a = 0.1\ m\ s^{-2}$, $t_1 = 240\ s$

Constant speed: $v = 24\ m\ s^{-1}$, $t_2 = 600\ s$

Deceleration to rest: $t_3 = 180\ s$



Distance = area under the graph.

$A = \frac{1}{2} \times 240 \times 24 = 2880\ m$

$B = 600 \times 24 = 14400\ m$

$C = \frac{1}{2} \times 180 \times 24 = 2160\ m$

Total distance, $s = 2880 + 14400 + 2160 = 19440\ m = 19.44\ km$

Tariff = 5000 shs per km

Charge for 19.44 km

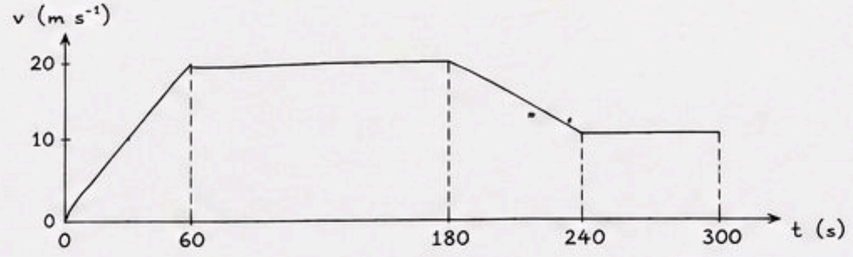
$= 19.44 \times 5000$

$= 97{,}200$ shs

Therefore, the ideal fare is Shs. 97,200. If he was charged more than this, the charge was NOT ideal.

ITEM 5

(a) The car moved according to the velocity–time graph shown below. Use it to determine the average speed for the whole journey and state whether the driver was penalized for exceeding $10\ m\ s^{-1}$.



Distance = area under the graph.

A (triangle) $= \frac{1}{2} \times 60 \times 20 = 600\ m$

B (rectangle) $= 120 \times 20 = 2400\ m$

C (trapezium) $= \frac{1}{2}(20 + 10) \times 60 = 900\ m$

D (rectangle) $= 60 \times 10 = 600\ m$

Total distance $= 600 + 2400 + 900 + 600 = 4500\ m$

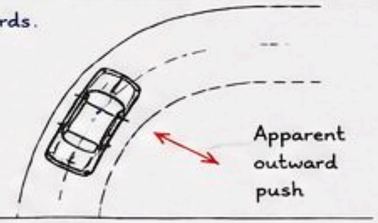
Total time $= 300\ s$

Average speed $= \frac{\text{total distance}}{} = \frac{4500}{300} = 15.0\ m\ s^{-1}$

Since $15.0\ m\ s^{-1} > 10\ m\ s^{-1}$, the driver was PENALIZED for exceeding the limit.

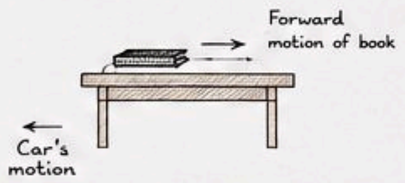
(b) As the car turned a sharp corner, the passenger felt pushed outwards. Give the reason for this.

- The car changes direction, but the passenger's body tends to move in a straight line (inertia).
- This produces an apparent outward force (centrifugal effect).



(c) When the car brakes suddenly, books on the table slide forward. Explain why this happens.

- When the car slows down, the books tend to continue moving forward due to inertia.
- The friction between the books and the table is not enough to stop them instantly, so they slide forward.



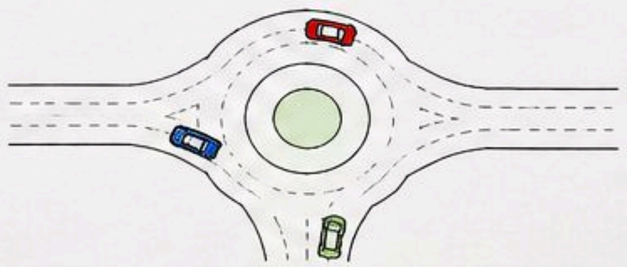
(d) Why are the tyres of the car made rough or patterned?

- Rough tyres increase friction between the tyres and the road.
- This provides better grip.
- It improves braking, reduces skidding and increases safety.



(e) The car moved in a roundabout as shown. State two reasons why the outer lanes are wider than the inner lanes.

1. Cars in the outer lane travel a longer distance in the same time, so a wider lane is needed for safety.
2. Wider lanes allow larger vehicles to turn easily without encroaching on other lanes.

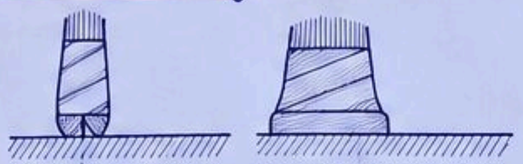


WENSSEC – ITEMS 4 & 5 (Physics)

ITEM 4

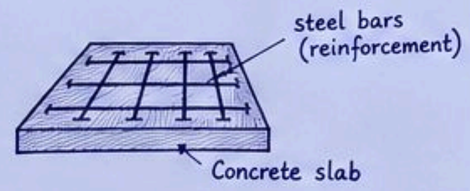
(a) Why was it difficult to walk on the wet concrete using sharp-pointed shoes?

- Pressure, $P = \frac{F}{A}$.
- Sharp-pointed shoes have a very small area of contact with the wet concrete.
- For the same weight (force), the pressure is very large, so the shoes sink into the wet concrete, making walking difficult.



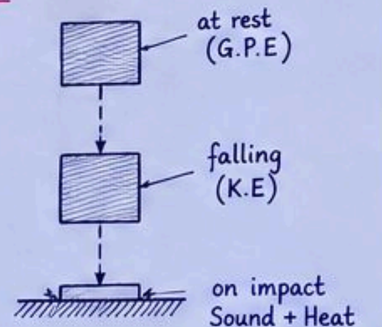
(b) Why did they reinforce the concrete slab on the first floor?

- The first floor slab must support its own weight, people, furniture and other loads.
- Steel reinforcement (steel bars) are provided to:
  - Increase the strength of the slab.
  - Resist tensile forces.
  - Prevent cracking.
  - Improve durability and safety of the building.



(c) Why did the blocks make noise when they hit the ground?

- As the blocks fall, their gravitational potential energy (G.P.E) is converted into kinetic energy (K.E).
- On striking the ground, the K.E is converted into sound energy, heat energy and some deformation energy of the blocks and the ground.
- The sound heard is due to the vibrations of the blocks and the ground.



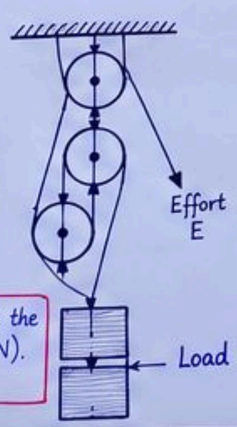
(d) Did the pulley system ease their work?

Given: Number of pulleys = 3 | Mass of each block = 6 kg
Number of blocks = 2 | Loss in efficiency = 18 %
$g = 10\ N\ kg^{-1}$

Load, $L = 2 \times 6 \times 10 = 120\ N$

I.M.A = number of supporting strands = 3

Efficiency, $\eta = 100\% - 18\% = 82\%$



$$\text{Actual M.A} = \frac{\eta}{100} \times \text{I.M.A}$$
$$= \frac{82}{100} \times 3$$
$$= 2.46$$

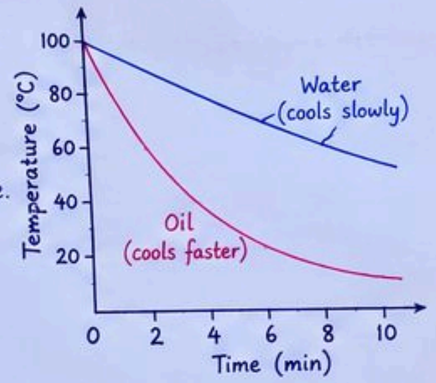
$$\text{Effort, } E = \frac{\text{Load}}{\text{M.A}}$$
$$= \frac{120}{2.46}$$
$$= 48.8\ N$$

Conclusion: Yes, the pulley system eased their work because the effort required (48.8 N) is much less than the load (120 N).

ITEM 5

(a) Why does cooking oil cool faster than water?

- Cooking oil has a lower specific heat capacity than water.
- It therefore requires less heat energy to change its temperature.
- Hence oil loses heat more quickly and its temperature falls faster than that of water.

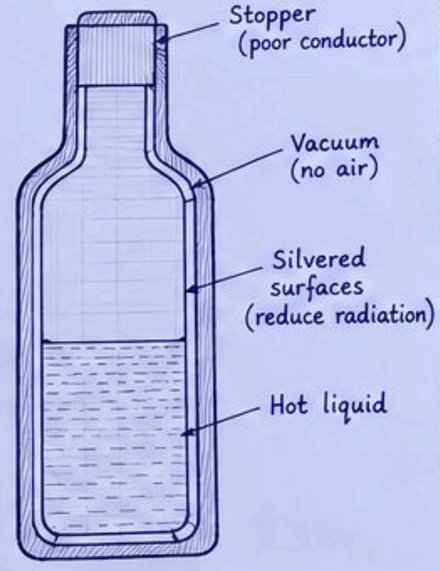


(b) How does a vacuum flask keep liquids hot for a long time?

A vacuum flask reduces heat loss by:

- The vacuum between the inner and outer walls, which prevents heat loss by conduction and convection.
- The inner surfaces are silvered (shiny) to reduce heat loss by radiation.
- The stopper is a poor conductor (usually plastic/cork) and prevents convection through the neck.
- The flask has a narrow neck to reduce heat loss.

Hence the liquid remains hot for a long time.



Key concepts tested:

- Pressure ($P = \frac{F}{A}$)
- Reinforced concrete
- Energy transformations
- Pulleys (Mechanical Advantage)
- Efficiency of machines
- Specific heat capacity
- Heat loss (conduction, convection, radiation)
- Vacuum flask

MASAKA

ITEM 4

(a) How many litres of water was pumped into the tank?

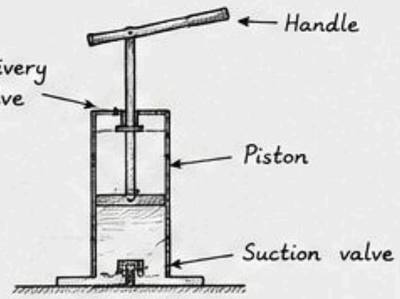
Given:
Power, $P = 3500\ W$
Time, $t = 8$ minutes $= 8 \times 60 = 480\ s$
Height, $h = 15\ m$
$g = 10\ N\ kg^{-1}$

Solution:
Energy supplied by the pump,
$E = P \times t = 3500 \times 480$
$E = 1.68 \times 10^6\ J$

This energy is used to lift water to a height h.
$E = mgh$
$1.68 \times 10^6 = m \times 10 \times 15$
$m = \frac{1.68 \times 10^6}{150}$
$m = 11200\ kg$

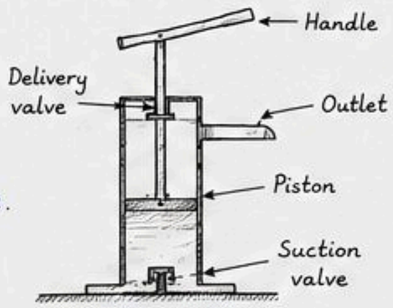
Mass of water = 11200 kg
Since 1 litre of water ≈ 1 kg
Volume of water = 11200 litres.

(b) Why is a lift pump not suitable for pumping water from this tank?
- A lift pump can raise water only up to a maximum of about 10 m.
- The tank is 15 m deep which is more than 10 m.
- Therefore a lift pump cannot raise the water to the surface effectively.



Lift pump

(c) How does a force pump solve the problem?
- A force pump uses a piston and delivery valve.
- When the handle is pressed down, water is forced into the cylinder through the suction valve.
- When the handle is lifted up, the delivery valve opens and water is forced out through the outlet.
- This produces pressure and can raise water to heights greater than 10 m.



Force pump

(d) Why did the cold metal feel colder than the wood?
- Metal is a good conductor of heat.
- It removes heat from our hand quickly.
- Therefore it feels colder.
- Wood is a poor conductor of heat.
- It removes heat slowly, so it feels warmer.

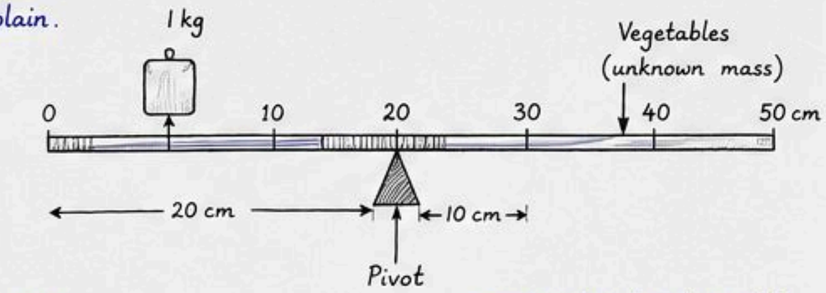
CONDUCTIVITY OF MATERIALS

Good conductors (e.g. metal)	Poor conductors (e.g. wood)
	
Heat flows quickly from the hand. Feels cold.	Heat flows slowly from the hand. Feels warm.

ITEM 5

(a) Is the weighing scale fair? Explain.

Given:
Length of beam = 50 cm
Pivot at 20 cm mark
Mass placed on one side = 1 kg
Mass of beam = 100 g = 0.1 kg
$g = 10\ N\ kg^{-1}$



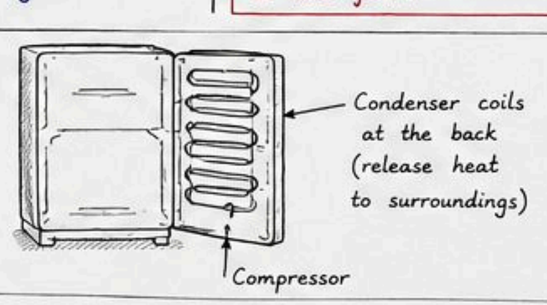
Weight of 1 kg mass, $W_1 = 1 \times 10 = 10\ N$
Weight of beam, $W_b = 0.1 \times 10 = 1\ N$
Centre of gravity of uniform beam is at middle = 25 cm mark.
Distance of beam's weight from pivot = $25 - 20 = 5$ cm

For the scale to be balanced, anticlockwise moment = clockwise moment.
Let W be the weight of vegetables placed at 30 cm mark.
$W \times 10 = 205$
$W = \frac{205}{10} = 20.5\ N$
Mass of vegetables, $m = \frac{W}{g} = \frac{20.5}{10} = 2.05\ kg$

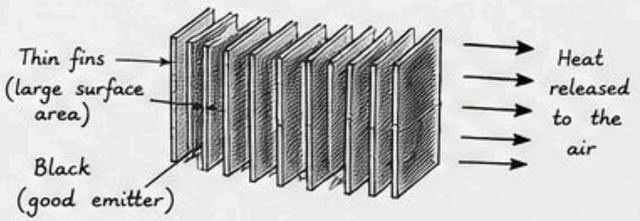
Moment due beam's weight $= 1 \times 5 = 5\ Ncm$
Moment due to 1 kg mass $= 10 \times 20 = 200\ Ncm$
Total clockwise moment $= 200 + 5 = 205\ Ncm$

The scale will balance with 2.05 kg not 1 kg. Therefore the scale is not fair.

(b) How does a refrigerator preserve milk?
- A refrigerator removes heat from the milk and transfers it to the surroundings.
- The temperature inside is kept low.
- Bacteria and microorganisms grow slowly at low temperatures.
- Therefore the milk does not spoil quickly.



(c) Why are the black fins at the back of the refrigerator thin and numerous?
- They provide a large surface area.
- A larger surface area allows more heat to be given out to the surroundings.
- They are painted black because black surfaces are good emitters of heat.
- Therefore heat is removed quickly and the refrigerator cools efficiently.



SUMMARY OF KEY IDEAS (MASAKA ITEMS 4 & 5)

ITEM 4 – MAIN IDEAS
- Work, Energy and Power: $P = \frac{W}{t}$, $E = mgh$
- Lift pump works only up to about 10 m.
- Force pump can raise water to greater heights.
- Good conductors (like metals) feel colder because they remove heat quickly.

ITEM 5 – MAIN IDEAS
- Moments: Moment = Force × Perpendicular distance
- A weighing scale is not fair if the beam's weight is not taken into account.
- Refrigeration works by removing heat and keeping the temperature low.
- Thin, black, numerous fins increase heat loss.

WAKISSHA

ITEM 6

(a) Help solve the client's confusion.

Electrical power generated at the waterfall is transmitted at HIGH VOLTAGE and LOW CURRENT.
This is because;

$$P = VI \quad \Rightarrow \quad I = \frac{P}{V}$$

For a given power, increasing voltage (V) reduces current (I).
Since power loss in transmission lines is;

$$P_{loss} = I^2R$$

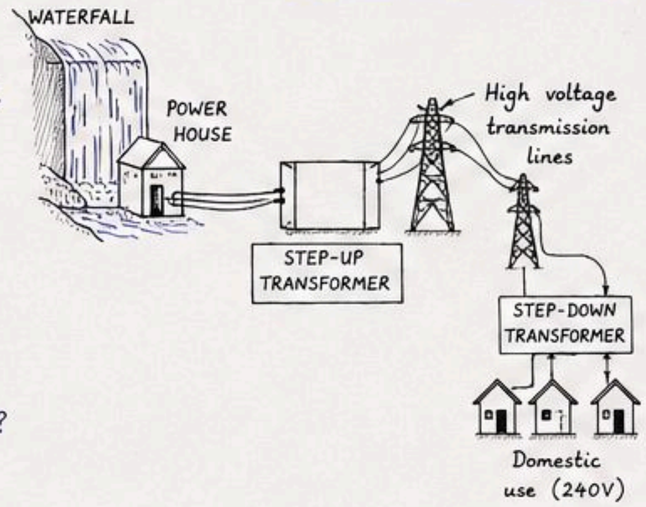
a smaller current greatly reduces energy loss.

(b) How would the procured item solve the problem?

The required item is a TRANSFORMER.

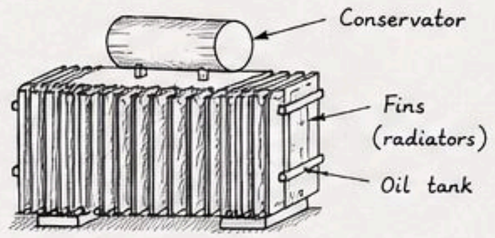
- It changes AC voltage.
- It steps up the voltage before transmission.
- It steps down the voltage for domestic use.
- This reduces transmission losses.

Hence electricity is transmitted efficiently over long distances and safely supplied to homes.



(c) Why do transformer coils have many fins?

- The fins increase the surface area.
- This allows faster loss of heat to the surroundings by convection and radiation.
- It prevents overheating and improves efficiency and life of the transformer.



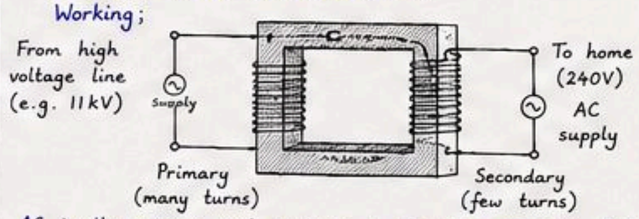
(d) Domestic wiring colour codes.

WIRE	COLOUR
Live (L)	Brown
Neutral (N)	Blue
Earth (E)	Green / Yellow

(e) Device installed before home use.

A STEP-DOWN TRANSFORMER.

Working;



AC in the primary coil produces a changing magnetic field.
This induces AC in the secondary coil.
Since secondary has fewer turns, voltage is reduced to a safe value (240V).

(f) Percentage power loss in transmission lines.

Given; Power, $P = 50{,}000$ W Voltage, $V = 500$ V
Resistance of each line, $R = 10\ \Omega$ Number of lines = 4

Current in the line; $I = \frac{P}{V} = \frac{50{,}000}{500} = 100$ A

Total resistance, $R_{total} = 4 \times 10 = 40\ \Omega$

Power loss, $P_{loss} = I^2R = (100)^2 \times 40 = 400{,}000$ W

Percentage power loss $= \frac{P_{loss}}{P} \times 100\% = \frac{400{,}000}{50{,}000} \times 100\% = \underline{\underline{800\%}}$

(g) Ways of reducing transmission losses.

- Transmit at very high voltages.
- Use thick conductors (low resistance).
- Use good quality conductors (low resistivity).
- Keep transmission lines as short as possible.
- Ensure proper maintenance of lines and joints.
- Using transformers to step up voltage.

ITEM 7

(a) Was the businessman cheated?

Cost per unit = 750 shs Number of days = 30
Calculate monthly energy consumption of each appliance.

- **Two electric irons**
  Power = 2 x 1200 W = 2400 W = 2.4 kW
  Used 2 h/day
  Energy = 2.4 x 2 = 4.8 kWh/day
- **Washing machine**
  Power = 2200 W = 2.2 kW
  Used 3 h/day
  Energy = 2.2 x 3 = 6.6 kWh/day
- **Three bulbs**
  Power = 3 x 100 W = 300 W = 0.3 kW
  Used 10 h/day
  Energy = 0.3 x 10 = 3 kWh/day

Appliance	Energy per day (kWh)	Energy per month (kWh)
Two irons	4.8	4.8 x 30 = 144
Washing machine	6.6	6.6 x 30 = 198
Three bulbs	3.0	3.0 x 30 = 90
Total	14.4	432 kWh

Monthly energy consumption = 432 kWh
Cost = 432 x 750 = 324,000 shs
Actual bill = 300,168 shs

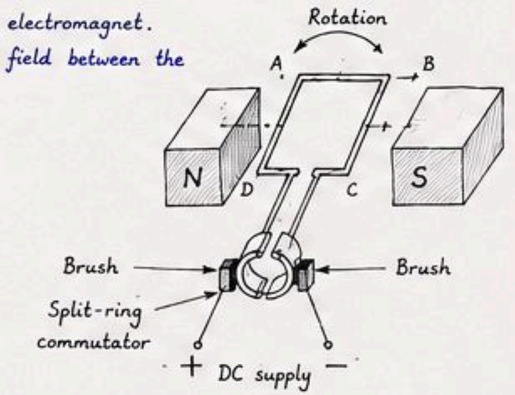
Since calculated cost (324,000 shs) is greater than the bill (300,168 shs), the businessman was NOT cheated.

(b) How does the rotating device work?

The rotating device in the washing machine is an ELECTRIC MOTOR.

Working;

1. When current flows through the coil ABCD, it becomes an electromagnet.
2. The magnetic field of the coil interacts with the magnetic field between the N and S poles.
3. A turning force (torque) acts on the coil.
4. The coil turns.
5. The split-ring commutator reverses the current in the coil every half turn.
6. The turning force continues in the same direction and the coil keeps rotating.



(c) How can electricity costs be minimized?

- Switch off appliances when not in use.
- Use energy saving devices e.g. LED bulbs.
- Reduce the time of using appliances (irons, washing machines).
- Do not leave appliances on standby.
- Maintain appliances regularly for efficiency.
- Use appliances of appropriate sizes and power.
- Take advantage of natural lighting and ventilation.

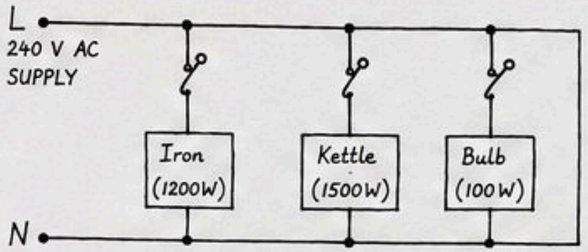
SAVE ENERGY
SAVE MONEY
☺

JJEB

JJEB – ITEM 6

(a) How should the house be wired?

- The house should be wired in parallel.
- In parallel, each appliance gets the full supply voltage (240 V).
- Each appliance works independently.
- If one appliance fails, others continue to work.
- It is safer and more convenient.



(b) Why is the bill not the same every month?

- The number of units (kWh) used changes from month to month.
- Different appliances may be used for longer or shorter times.
- Seasonal changes affect usage (e.g. more lights in dark months).
- Increase or decrease in the number of people at home.
- Some appliances may be added or removed.

Hence energy consumption varies, so the bill also changes.

(c) Name TWO safety components in the installation.

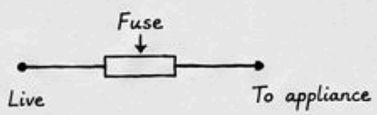
1. Fuse
2. Earth wire

(d) Why are safety components important?

- They protect people from electric shock.
- They prevent damage to appliances.
- They prevent fires due to overcurrent or short circuits.
- They make the installation safe.

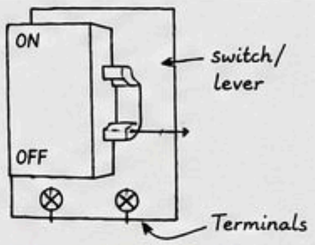
(e) How does a fuse protect the circuits?

- A fuse is a thin wire of low melting point.
- If current becomes too large (overload or short circuit), the fuse heats up.
- It melts (blows), breaking the circuit.
- This stops the flow of current.
- Therefore appliances and wires are protected from getting damaged or causing fire.



(f) How do circuit breakers protect the circuit?

- They automatically break the circuit when current is too high.
- This may be due to overload or short circuit.
- They have a bimetallic strip or an electromagnet.
- The strip bends or the electromagnet pulls a latch.
- The switch opens and stops the current.
- The circuit can be reset after the fault is removed.



SUMMARY – ITEM 6 (JJEB)

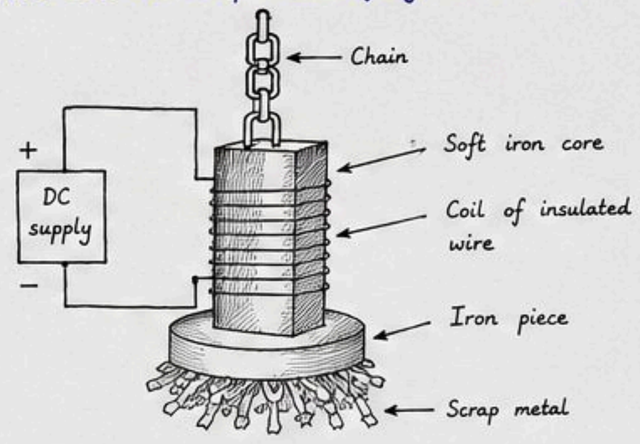
- House wiring (parallel)
- Electricity bills
- Safety components (fuse, circuit breaker, earth wire)
- Protection from overload and short circuit
- Importance of safety in electrical installations

JJEB – ITEM 7

(a) What is an electromagnet?

An electromagnet is a temporary magnet produced when electric current passes through a coil of wire, usually wound on a soft iron core.

(b) How does the scrap metal lifting device work?



Working:

1. When current flows through the coil, the soft iron core becomes magnetized.
2. The electromagnet attracts the scrap metal.
3. When the current is switched off, the core is demagnetized.
4. The scrap metal falls down.

(c) State TWO uses of electromagnets.

1. Lifting scrap metals.
2. Electric bells / relays.

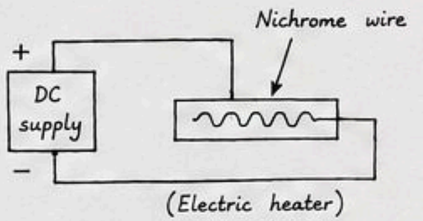
(Any other correct use)

(d) Why is the core made of soft iron?

- Soft iron is easily magnetized.
- It loses magnetism quickly when current is switched off.
- Therefore the scrap metal is released easily.

(e) How does the wire of the electric heater heat up?

- When electric current flows through the nichrome wire, it offers resistance.
- Electrical energy is converted into heat energy.
- This is called the heating effect of electric current.
- The wire becomes hot and heats the surroundings.



(f) State TWO applications of the heating effect of current.

1. Electric cookers.
2. Electric irons.

(Any other correct application)

SUMMARY – ITEM 7 (JJEB)

- Electromagnet: definition and uses
- Lifting device (electromagnet)
- Soft iron core
- Heating effect of current
- Applications of heating effect

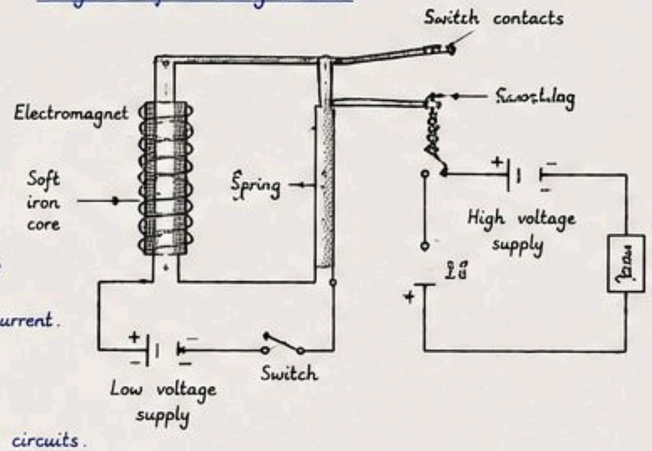
KAMSSA

ITEM 6

(a) What is a relay switch?

A relay switch is an electrically operated switch. It uses an electromagnet to open or close a circuit in another part of the circuit.

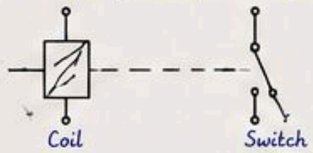
Diagram of a relay switch



(b) Why are relay switches used in large circuits?

- They allow a small current to control a large current.
- They increase safety.
- They reduce wear and tear on switches.
- They allow automation and remote control of circuits.
- They isolate control and load circuits.

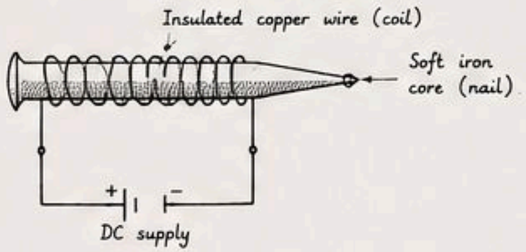
(c) Draw the symbol of a relay switch.



(d) State TWO uses of relay switches.

1. In car horns.
2. In automatic street lights.
- (Any other correct use)

(e) Draw a labelled diagram of an electromagnet.



(f) Why does an electromagnet lose its magnetism when the current is switched off?

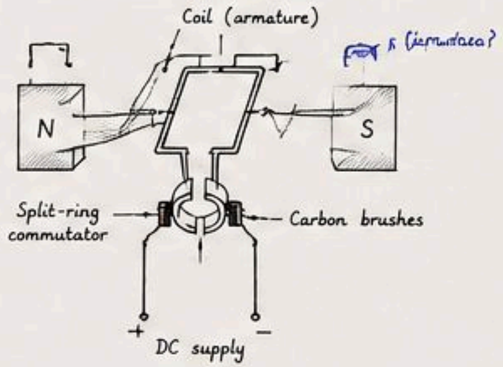
Because an electromagnet is a temporary magnet. When the current is switched off, the magnetic field disappears and the soft iron core is no longer magnetized.

(g) State TWO uses of electromagnets.

1. In electric bells.
2. In lifting cranes to lift scrap metal.

(Any other correct use)

(h) Draw a simple electric motor and label the parts.



(i) State TWO uses of electric motors.

1. In electric fans.
2. In water pumps.
- (Any other correct use)

SUMMARY – ITEM 6 (KAMSSA)

- Relay switch – definition, diagram, uses.
- Symbols of relay switches.
- Electromagnets – diagram, properties, uses.
- Electric motors – diagram, working, uses.
- Applications of magnetism in daily life.

ITEM 7

(a) Explain the term electrical energy.

Electrical energy is the energy supplied or used by an electric current when it flows through a conductor.

Important formula

Electrical energy, $E = P \times t$

where: E = energy (J)
P = power (W)
t = time (s)

(b) How much electrical energy is used by a bulb rated 100 W if it is used for 5 hours?

Power, $P = 100$ W

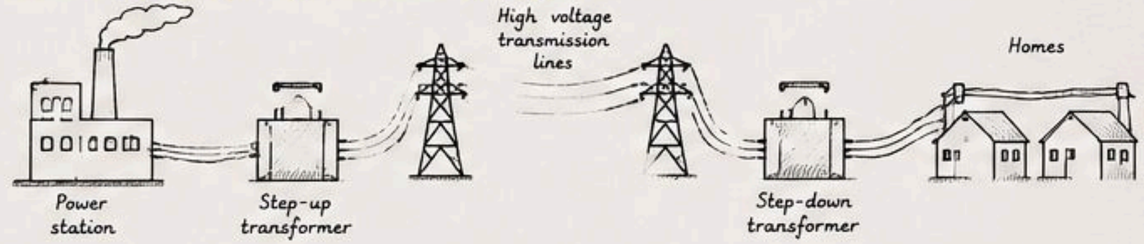
Time, $t = 5$ hours $= 5 \times 3600 = 18{,}000$ s

Electrical energy, $E = P \times t$
$= 100 \times 18{,}000$
$= 1{,}800{,}000$ J (1.8 MJ).

(c) Name the system used to supply electricity from the power station to homes.

National grid system.

(d) Draw a simple diagram of the national grid system.



(e) Why is electricity transmitted at high voltage?

- To reduce the current in the transmission lines.
- This reduces power loss (I^2R losses).
- Hence electricity is transmitted efficiently over long distances.

(f) Why is a step-up transformer used at the power station?

To increase the voltage to a very high value so that current is reduced and power loss in transmission lines is minimized.

(g) Why is a step-down transformer used near homes?

- To reduce the voltage to a safe value (about 240 V).
- This makes it safe for domestic use.

(h) State TWO advantages of hydropower.

1. It is a renewable source of energy.
2. It does not produce air pollution.
- (Any other correct advantage)

(i) State TWO disadvantages of hydropower.

- (Any other correct disadvantage)

SUMMARY – ITEM 7 (KAMSSA)

- Electrical energy – definition, calculation.
- National grid system – diagram and description.
- Transformers in transmission.
- Reasons for using high voltage.
- Advantages and disadvantages of hydroelectric power.
- Electrical energy in daily life.

Energy saved is money saved! ☺

MASAKA ITEM 6

In a hospital ward, two fans rated 240 V, 800 W each are powered by d.c motors. They operate for six hours daily. Electricity costs 950 sh per unit (1 kWh).

(a) (ii) Find the cost of powering each fan for one day.

Power of one fan, P = 800 W = 0.8 kW

Time of operation per day, t = 6 h

Energy used per fan per day,

$E = P \times t = 0.8\ kW \times 6\ h = 4.8\ kWh$

Cost per fan per day = 4.8×950

= **4,560 shs**

Therefore, each fan costs 4,560 shs per day.

For both fans in one day,

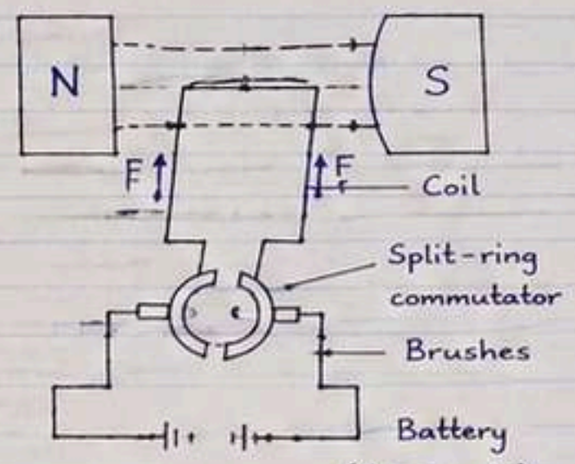
Total cost = $2 \times 4,560 = 9,120$ shs

(b) How does the d.c motor in the fan work?

When current flows through the coil of the motor, the coil becomes an electromagnet. The two sides of the coil which are in the magnetic field experience forces in opposite directions. These forces produce a turning effect (torque) on the coil.

The split-ring commutator reverses the direction of current in the coil after every half turn. This keeps the turning effect in the same direction. The coil continues to rotate.

The coil is connected to the fan blades through the shaft. Therefore, as the coil rotates, the shaft and the blades also rotate and the fan blows air.



(c) How can the problems with the fans be solved?

Problem 1: Low speed

The speed of a d.c motor can be increased by:

- Increasing the current through the coil (without exceeding the safe limit).
- Increasing the strength of the magnetic field.
- Increasing the number of turns of the coil.

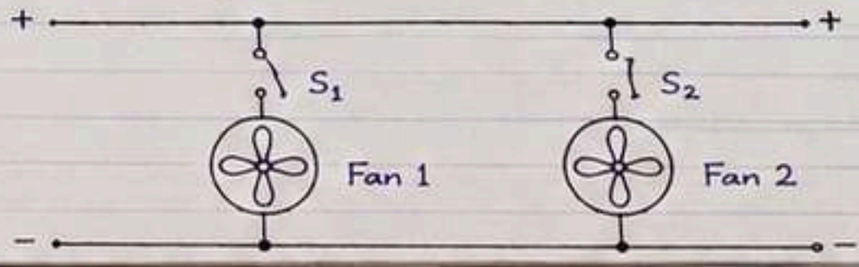
Problem 2: Both fans work together

The fans should be connected in parallel.

Each fan should be controlled by its own switch.

This allows one fan to operate without the other.

Therefore, when cooling is required in only one room, the second fan can be switched off to save electricity.



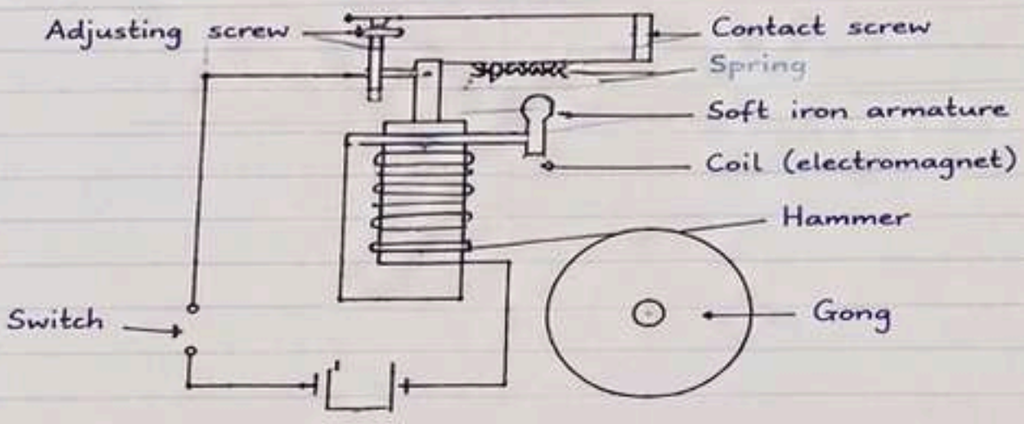
MASAKA ITEM 7

An electric bell is supplied with current by pressing a switch. The soft iron armature is attracted by a magnet formed by a coil. Changing the small current resistance of the electric bell from 24 Ω to 50 Ω reduces the speed at which it normally rings. The bell is to be protected against power surges.

(a) How does the electric bell solve the problem?

When the switch is pressed, current flows through the coil. The coil becomes an electromagnet and attracts the soft iron armature. The armature moves and the hammer strikes the gong to produce sound.

As the armature moves, it breaks the contact screw. The circuit is opened and current stops flowing. The electromagnet then loses its magnetism and the armature is pulled back by the spring. The contact is made again and current flows. This process is repeated rapidly. Therefore, the hammer strikes the gong continuously and a loud ringing sound is produced which can be heard at the gate.



(b) (ii) What would be the effect on the bell if the soft iron armature is replaced with steel?

The bell would not operate properly or might even stop working. This is because steel is hard to magnetise and once magnetised it retains its magnetism for a long time. Therefore the armature would remain magnetised and would not be easily released when the current stops. The armature would not return quickly to its original position, so the contact would not be made and broken rapidly. Hence the bell would not ring continuously.

(ii) What effect would there be on the bell if the small current resistance of the electric bell is changed from 24 Ω to 50 Ω?

Using Ohm's law, $I = V/R$.

For the same supply voltage, increasing resistance from 24 Ω to 50 Ω reduces the current through the coil.

A smaller current makes the electromagnet weaker.

The magnetic pull on the armature becomes less.

Therefore the armature will not be attracted strongly and will move more slowly. The hammer will strike the gong with less force and the bell will ring more slowly or may not ring loudly.

(c) How can the bell be protected against power surges?

The bell can be protected by:

- Installing a fuse of the correct rating in series with the bell.
- Using a circuit breaker.
- Using a surge protector (surge arrester).
- Proper earthing of the electrical installation.
- Installing lightning protection for the building.

WENSSEC ITEM 6

(a) (ii) Why connecting wires become hot

When electric current flows through a wire, energy is dissipated in the form of heat due to the resistance of the wire.
The heat produced (H) is given by

$$H = I^2Rt$$

where I = current in amperes
R = resistance of the wire in ohms
t = time in seconds.

The connecting wires become hot because;
- When a large current flows,
- when the resistance of the wire is high,
- when current flows for a long time.

Overloading causes excessive current to flow, which produces a lot of heat that may melt the insulation and cause a fire.

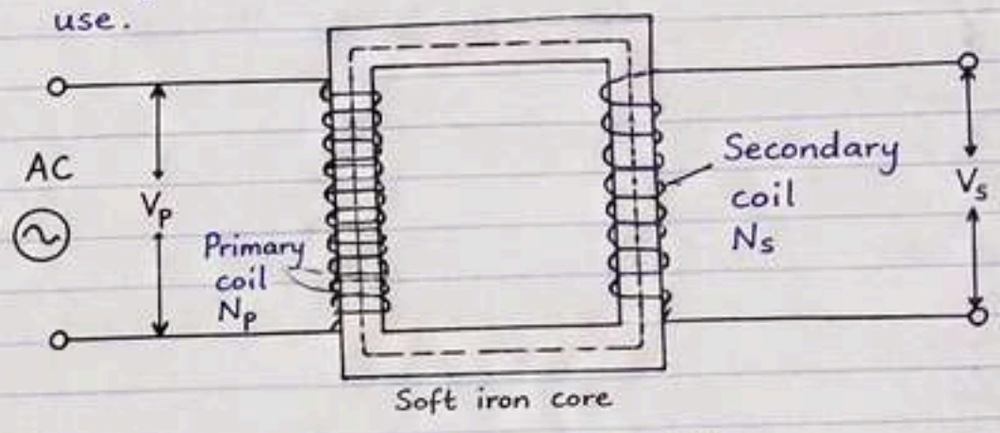
(ii) Meaning of 7.5 A, 240 V

240 V means the fridge is designed to operate normally when connected across a potential difference of 240 volts.
7.5 A means that when operating at 240 V, the fridge draws a current of 7.5 amperes.

The power of the fridge; $P = VI$
$= 240 \times 7.5$
$= 1800\ W$
$= \underline{1.8\ kW}$

(b) Device used to lower the voltage

The device used to lower the voltage is a STEP-DOWN TRANSFORMER.
It works on the principle of electromagnetic induction. An alternating current in the primary coil produces a changing magnetic flux in the soft iron core. This changing flux links the secondary coil and induces an e.m.f in it.
For a step-down transformer,
Number of turns on secondary, $N_s < N_p$
Therefore $V_s < V_p$
It is used to reduce the high transmission voltage to a safe low voltage for domestic use.



(c) Cost of one unit of electricity

(i) Fridge

Power, P 1.8 kW
Time used = 4 hours per day x 15 days
= 60 h
Energy used, $E = P \times t = 1.8 \times 60 = 108\ kWh$

(ii) Eight bulbs

For one bulb; $R = 5\Omega$, $I = 4\ A$
Power, $P = I^2R = 4^2 \times 5 = 80\ W = 0.08\ kW$
For eight bulbs; $P = 8 \times 0.08 = 0.64\ kW$
Time used = 12 hours/day x 30 days = 360 h
Energy used, $E = P \times t = 0.64 \times 360 = 230.4\ kWh$

(iii) Three phones

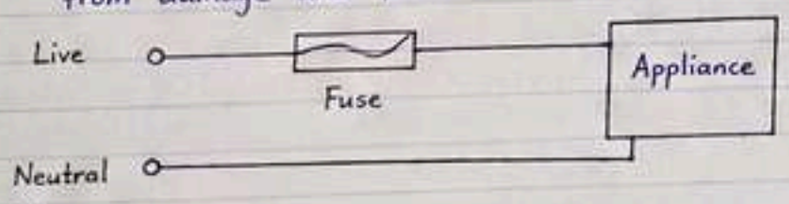
Current for one phone; $I = 4500\ mA = 4.5\ A$
Power for one phone, $P = VI = 240 \times 4.5$
$= 1080\ W = \underline{1.08\ kW}$
For three phones; $P = 3 \times 1.08 = 3.24\ kW$
Time used = 1.5 hours/day x 30 days = 45 h
Energy used, $E = P \times t = 3.24 \times 45 = 145.8\ kWh$

Total energy used;
$E_{total} = 108 + 230.4 + 145.8$
$= 484.2\ kWh$;
Total cost = UGX 121,050
Cost per unit (1 kWh) = $\frac{121,050}{484.2}$
$= \boxed{UGX\ 250}$

(d) Any two house-wiring components and their importance.

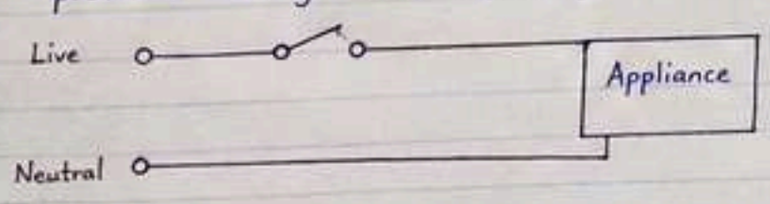
(i) Fuse

A fuse is a safety device connected in series with the live wire. When the current exceeds the safe value, the fuse wire melts and breaks the circuit, thus protecting the appliances and wiring from damage and fire.



(ii) Switch

A switch is used to open or close a circuit. It allows the user to control the flow of current to an appliance. It helps in saving energy and also provides safety when repairing.



WENSSEC ITEM 7

(a) What probably happened

The fire could have been caused by an electrical fault such as excessive current due to overloading or short circuit. When too much current flows through the connecting wires, a lot of heat is produced ($H = I^2Rt$). This heat can melt the insulation of the wires and ignite the surrounding materials causing fire.
Electrical installations should therefore be properly protected using fuses or circuit breakers, correct wiring and proper earthing.

(b) The resistance required

Given; Power of the device, P = 108 W
Voltage, V = 240 V
Normal current, $I = \frac{P}{V} = \frac{108}{240} = 0.45\ A$
The resistor should allow a current not exceeding 13 A.
Using Ohm's law, $R = V\ \frac{I}{I_{max}}$
$R = \frac{240}{13} = 18.46\ \Omega$
$\therefore$ The resistance required $\approx \underline{18.5\ \Omega}$

(c) How the device achieves its intended purpose

The device is rated 108 W, 240 V. This means that when it is supplied with 240 V, it consumes 108 W of electrical power. The resistor is connected in the circuit to limit the current so that it does not exceed the safe value (13 A). By limiting the current, the device works safely and does not get damaged by overheating. Electrical energy is thus converted into the intended useful form.

(d) How the new building can be protected

- Install a lightning conductor to provide a safe path for lightning charge to the ground.
- Properly earth the electrical installation.
- Use fuses or circuit breakers of the correct rating.
- Use correctly rated wires to avoid overheating.
- Avoid overloading sockets and circuits.
- Regularly inspect electrical wiring and appliances.

The iron and copper rods can be used for earthing and lightning protection.

